STAT 330-001

40142034 64

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Test: 641565 Form: A
Entry: 07:43pm 10/21/2026
Employee: Addie T.

Wadsworth, Spencer Gordon
Checked out:

SPECIAL INSTRUCTIONS

TC Calc
Notes: 3x5
Dictionary
Take a deep breath. Be careful to read instructions thoroughly. Answer questions completely.

Please keep this sheet **VISIBLE** on your desk at all times. When you are finished, *staple this sheet* to the front of your test booklet and return all materials to the Grading Table. You may use the space below for notes or scratch paper.

PLEASE TURN OFF YOUR CELL PHONE

NS

40142034

064

STAT 330 - Exam 1

Section 001 (Wadsworth), Section 002 (White)
October 16th - 21st, 2026

Instructions

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- Do not discuss any aspect of this exam with anyone except a Stat 330 TA or instructor until scores are posted.
- You are allowed to use a calculator for this exam.
- In Parts II and III, **make sure to clearly show your work for any calculations that you make and to give your answers in context of the data sets.** Place your final answer inside the box, if one is provided.
- There are 24 questions. Make sure you answer all of them.
- There is extra scratch paper at the end of the exam.

PART I

Question 1: (Circle one answer) Partial regression (or added-variable) plots illustrate the effect of adding an additional variable to the model after accounting for the variables that are already included in the model.

(A) TRUE

(B) FALSE

Question 2: (TRUE/FALSE) If the relationship between x and y is nonlinear, a simple linear regression model may still be feasible after transforming the data.

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Question 3: (Circle one answer) When adding an additional variable to a linear regression model, the model's R^2 ...

(A) always increases or remains unchanged.

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Question 4: (Circle all that apply) Which of the following statements describe possible effects of multicollinearity?

(A) It can cause the linearity assumption to be violated.

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(C) It can result in larger standard errors, and thus greater uncertainty in estimates.

(D) It can cause the coefficient of determination (R^2) to become negative.

Question 5: (Circle all that apply) Which of the following are true statements about multiple linear regression (MLR)?

(A) BIC will usually favor a model with more predictors than the AIC criterion when performing model selection.

(B) Cook's distance can be used to identify influential observations.

(C) Transformations of the x's or y may help alleviate problems with linearity, equal variance, or normality.

(D) The least-squares estimate of β can be obtained by fitting a simple linear regression for each x_k only.

(E) The assumptions on the error variables (i.e., the ϵ_i) in a multiple linear regression model are exactly the same as they are for simple linear regression.

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(C) CV ensures model assumptions are met which eliminates a time-consuming step of the modeling process.

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Question 7: (Circle all that apply) In datasets with high multicollinearity and/or redundant covariates, what advantages do LASSO and/or ridge regression have in comparison to the traditional ordinary least squares (OLS) approach to model fitting?

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(C) They produce more accurate confidence intervals and hypothesis tests

(D) Ridge regression often handles multicollinearity well

(E) They can handle datasets where the number of predictors exceeds the number of observations

(F) They can take advantage of the bias-variance tradeoff to produce more stable and accurate predictions

Question 8: Why are LASSO and ridge regression sometimes referred to as "shrinkage" methods? Specifically, what is being shrunk and how is this accomplished?

The number of predictors in the model can sometimes decrease or be reduced to zero entirely. This is accomplished by transformations.

Question 9: A multiple linear regression model with $p = 3$ predictors is fit using $n = 100$ observations. Observation i has standardized residual $r_i = 2.4$ and leverage $h_{ii} = 0.25$.

Use

$$D_i = \frac{r_i^2}{p+1}\left(\frac{h_{ii}}{1-h_{ii}}\right).$$

(Circle one answer) Which conclusion is most appropriate?

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(G) Cook's distance cannot be calculated without knowing the observation's original residual.

Question 10:

(A) What is the formula for $\widehat{\beta}$?

$\hat{\beta} = y(X^TX)^{-1}X$

(B) What is the conceptual difference between $\widehat{\beta}$ and β?

$\hat{\beta}$ is an estimate, while β is known.

(C) Explain how least squares can be used to estimate regression coefficients. You do not need to derive anything, but you should be clear about how this is done conceptually.

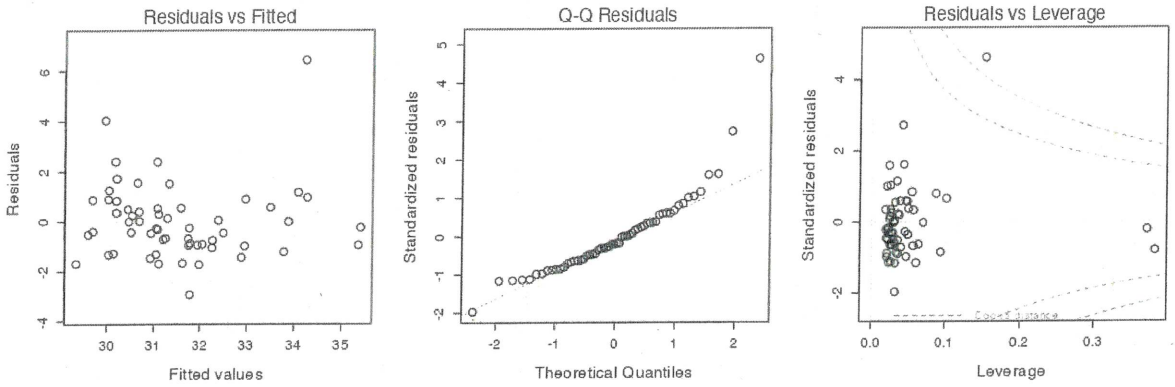
We choose a $\hat{\beta}$ value to minimize SSE and $\sum_i e_i^2$.

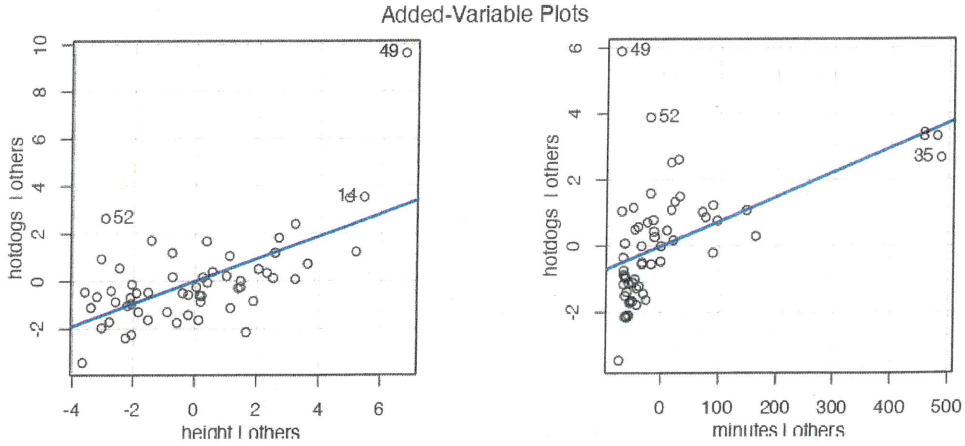
PART II

Some consider competitive eaters to be among the most inspiring individuals in modern times. We are interested in knowing what factors relate to a competitive eater's ability to consume. In a recent state fair, a contest was held where contestants competed to see who could eat the most hotdogs in 10 minutes. Data were collected on 58 amateur competitive eaters, most of whom were from the 4 largest counties in the state. For each eater, the total number of hotdogs consumed in a single 10 minute sitting was recorded. Each contestant's height in inches was also recorded, and each contestant was asked "How many minutes per week do you spend training for a contest?" The variables are `hotdogs`, `height`, and `minutes`. Model (1) is used for analysis. Use the plots and model fit output below to answer questions 11-13.

Model 1

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \texttt{minutes}_i + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{1}$$





Coefficient	Estimate	Std. Error	t value	$\Pr(>\|t\|)$
Intercept	-1.295392	5.467711	-0.237	0.813602
`height`	0.472892	0.080352	5.885	2.46e-07
`minutes`	0.007417	0.001871	3.963	0.000215

Other Summaries	Value
Breusch-Pagan p-value	0.071
Shapiro-Wilk p-value	2.43e-8
CV error	1.84623

Question 11: State the estimated value of β_1, and interpret its value in the context of the problem

The estimate is 47.2892 for height.

Holding minutes constant, each additional inch in height is associated with a 47.2892% increase in the conditional median of number of hotdogs.

Question 12: **(Circle all that apply)** Which of the following are true?

(A) The 95% confidence interval of β_2 does **not** contain 0

(B) The 95% confidence interval of β_2 does contain 0

(C) There is an 81.4% probability that $\beta_0 = 0$

(D) $\beta_1 = 0.4729$

Question 13: Given what you know, provide a brief assessment of how well the data satisfy each of the following multiple linear regression assumptions:

(A) The relationships between the explanatory and response variables are linear

Given the data and looking at the Fitted vs. Residual and Added Variable plots, the relationship of x and y is not very linear. Our Residuals vs. Fitted plots shows curvature in the points, but the AV plots are less obvious.

(B) There is independence between each observation

All the data was gathered from the same contest and most individuals were from the largest four counties in the state. There is potential for lack of independence given these data collection methods.

(C) The errors are normally distributed

The errors do not seem to be normally distributed. There is some skewness on both tails of the Q-Q plot. The deviation from the line suggests non-normality in the model.

(D) The errors have constant variance (homoscedasticity)

We have a Breusch-Pagan value close to zero which implies equal variance. The points on the Residuals vs. Fitted plot also appear to be primarily equally distributed.

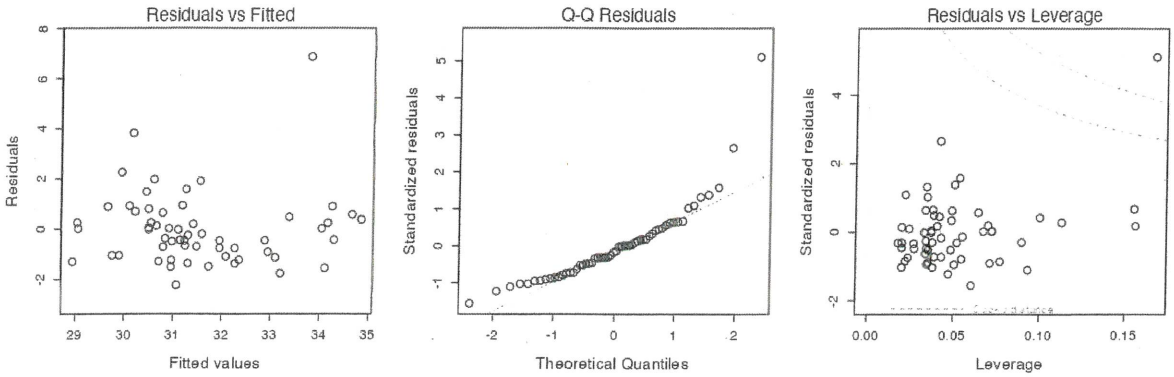
(E) There are no unduly influential points

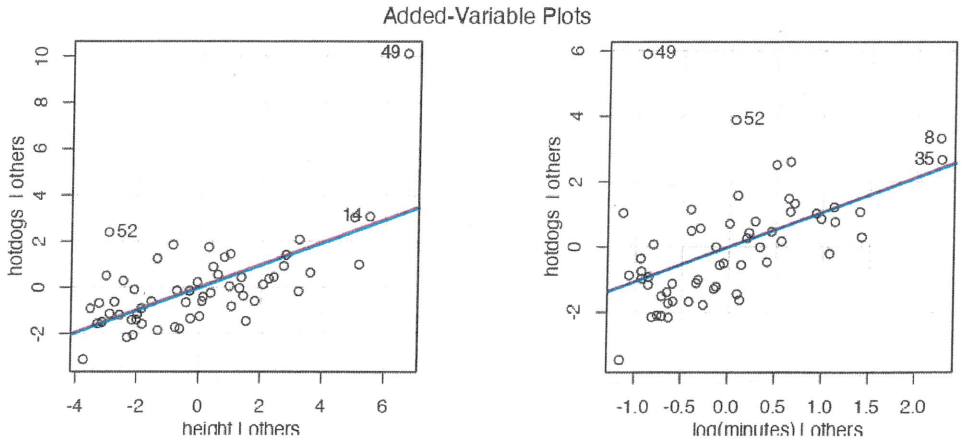
I am most concerned with the QQ Residuals plot and the skewness. additionally, the minutes I athers AV plot has some high x values that could be concern for leverage.

Model 2

Model (2) below represents another possible multiple linear regression model for the same data. Here the variable `minutes` was log-transformed prior to fitting the model. Use model (2) and the output below to answer questions 14-16

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \log(\texttt{minutes}_i) + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{2}$$





Coefficient	Estimate	Std. Error	t value	$\Pr(>\|t\|)$
Intercept	-6.05589	5.43620	-1.114	0.27
`height`	0.48834	0.07821	6.244	6.48e-08
log(`minutes`)	1.06702	0.24079	4.431	4.51e-05

Other Summaries	Value
Breusch-Pagan p-value	0.028
Shapiro-Wilk p-value	3.43e-5
CV RMSE	1.66702

Question 14: A Breusch-Pagan test for model (1) results in a p-value of 0.071 while a Breusch-Pagan test for model (2) results in a p-value of 0.028. A researcher concludes that model (1) is clearly superior to model (2). How would you respond?

The results for model 2 are significant while model 1 is not, BUT that does not automatically mean we should discard model 2. Small deviations may lead to significant test results. we should monitor it and continue performing tests.

Question 15: Explain what advantage(s) and disadvantage(s) model (2) provides over model (1)

Model 2 does have an advantage in its AV plots. This helps support the linearity assumption.

Some disadvantages are that it has a significant Breusch-Pagan p-value and larger t-values. This suggests more variation and a potential violation of the equal variance assumption.

Question 16: State the estimated value of β_2 and provide an interpretation in context

The estimated β_2 value is 0.02817 Holding height constant, a one log(minute) increase is associated with about a 2.817256% increase in the conditional median of number of hotdogs.

$\beta_2 = \log(1.06702)$

$\beta_2 = 0.02817256$

Use the following for questions 17-19

Your best friend reveals that he will be competing in the hotdog eating contest at the next state fair. You know that he is 68 inches tall and that he trains 79 minutes per week. Note that log(79) = 4.369. Use this information and the fit for model (2) to answer the following questions.

Question 17: Use model (2) to calculate the predicted number of hotdogs your friend can consume in 10 minutes

$e^{\log(79)} = 6.670048233$ hot dogs in 10 minutes.

Question 18: To create a confidence or prediction interval for the expected number of hotdogs your friend can consume in 10 minutes, what is the correct number of degrees of freedom for the critical t^* used to calculate the margin of error?

(A) 55

(B) 57

(C) 56

(D) 48

Question 19: You're given that $s\sqrt{x^*(X^\top X)^{-1}x^*} = 0.204$, $s\sqrt{1 + x^*(X^\top X)^{-1}x^*} = 1.485$, and $t^*_{df,0.975} = 2.004$. Give a 95% *prediction* interval for the number of hotdogs your friend can consume in 10 minutes.

$6.670048233 + 2.004(0.204) = 7.078864233$

$6.670048233 + 2.004(1.485) = 9.645988233$

$(7.078864233, 9.645988233)$

Part III - Application

Questions 20-24 concern the following dataset:

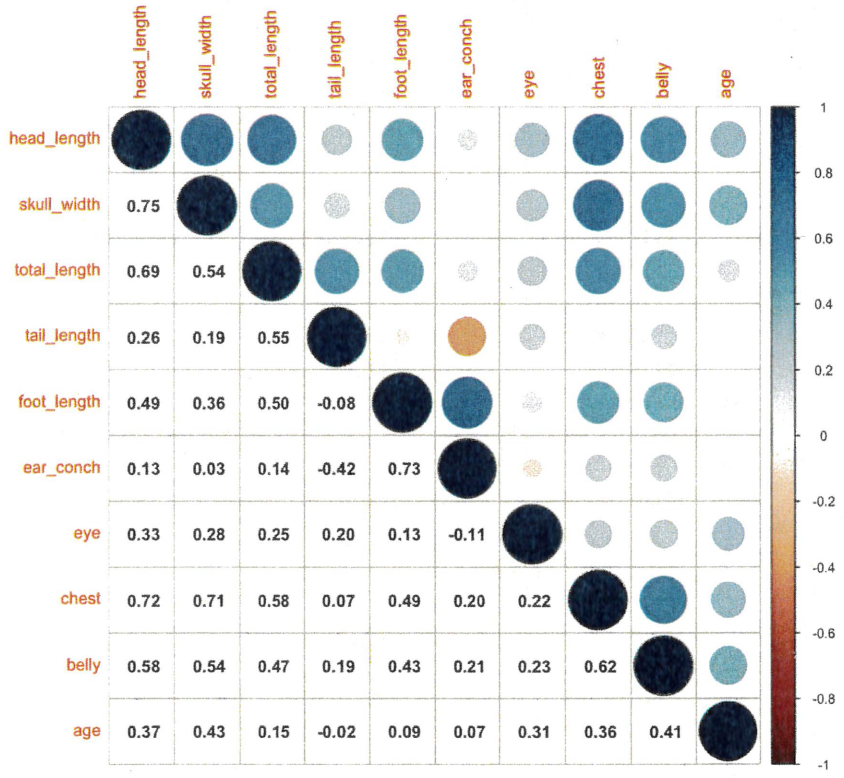
In 1995, Australian zoologists conducted a study of local possum populations. They trapped dozens of specimens, determined their ages, and took measurements of several physical attributes from each possum. The process of determining a wild animal's age can be challenging and imprecise, and usually requires a detailed analysis of the animal's teeth, skin, skeletal structure, or DNA. Among other questions, researchers were interested in whether a possum's age could be predicted using only easy to obtain physical measurements. For this analysis we will focus on the 59 male possums from the sample. The variables in the dataset are as follows (see below). A multiple linear regression model to predict age using all other variables was fit to these data, resulting in the following model summary and coefficient estimates (see below)

Variable	Description
head_length	in mm
skull_width	in mm
total_length	in cm
tail_length	in cm
foot_length	in mm
ear_conch	ear conch circumference (in mm)
eye	width of right eye (in mm)
chest	chest girth (in cm)
belly	belly girth (in cm)
age	in years

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -13.22377    7.34105  -1.801   0.0778 .
head_length   0.07077    0.10120   0.699   0.4877
skull_width   0.16559    0.10599   1.562   0.1247
total_length -0.04752    0.10167  -0.467   0.6423
tail_length  -0.05049    0.17030  -0.296   0.7681
foot_length  -0.21499    0.10733  -2.003   0.0507 .
ear_conch     0.15730    0.10231   1.538   0.1306
eye           0.43736    0.22549   1.940   0.0582 .
chest         0.01883    0.18154   0.104   0.9178
belly         0.20983    0.11580   1.812   0.0761 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.655 on 49 degrees of freedom
Multiple R-squared:  0.3538,    Adjusted R-squared:  0.2352
F-statistic: 2.982 on 9 and 49 DF,  p-value: 0.006552
```

A correlation plot (shown below) of these variables points towards some level of multicollinearity in the data. The maximum VIF among these variables is approximately 4 with an average VIF of approximately 3.



Question 20:

(a) Using the information from the summary table above, produce a 95% confidence interval for the coefficient on `skull_width`. The 0.975 critical value of a t-distribution with 49 degrees of freedom is 2.01. You may report your answer in the form $a \pm b * c$.

$0.16559 \pm 2.01(0.10599)$

$+: 0.3786299$

$-: -0.0474499$

$(-0.047, 0.379)$

(b) A 95% confidence interval for $\beta_{\texttt{belly}}$, the coefficient on belly width, is (-0.023, 0.443). Interpret this interval within the context of the data.

We are 95% confident that the true value of belly width is between (-0.023, 0.443). If we took infinitely more samples, we would expect 95% of them to contain the true population value.

Question 21: Note from the model summary that while the overall F-test of model significance produces a p-value of 0.0065, none of the individual model coefficients are found to have a significant effect at the $\alpha = 0.05$ level. What do you think is driving this "contradiction"? (It's not actually a contradiction.)

All of the predictors could be related to each other through multicollinearity. Individually, none of them are significant, but they have a significant effect together.

Question 22: Briefly describe how the "best subsets" algorithm using AIC as its criterion performs variable selection.

Best subsets is the model search algorithm and AIC is the model selection method. Together, they decide what model best fits the data.

Question 23: Best subsets using AIC as a criterion identified the following model with five variables:
`lm(age ~ skull_width + foot_length + ear_conch + eye + belly, data = possum)`
This model produces the following summary and coefficient estimates:

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -14.76661    4.97828  -2.966  0.00451 **
skull_width   0.20221    0.07990   2.531  0.01439 *
foot_length  -0.22936    0.09385  -2.444  0.01789 *
ear_conch     0.18239    0.08653   2.108  0.03979 *
eye           0.45001    0.21687   2.075  0.04285 *
belly         0.21405    0.10316   2.075  0.04287 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.609 on 53 degrees of freedom
Multiple R-squared:  0.3396,    Adjusted R-squared:  0.2773
F-statistic:  5.45 on 5 and 53 DF,  p-value: 0.0004056
```

Is this model guaranteed to have the highest adjusted-R^2 among all possible models.

(A) TRUE

(B) FALSE

Question 24: Suppose we want to compare the out-of-sample predictive performance of the full model and the model chosen by best subsets, and ultimately select the model with the smaller cross-validated RMSE.

Cross-validation using repeated random training-test splits can be structured around a for loop over some number of iterations (let's say 100), with some setup happening prior to the start of the loop, the bulk of the code within the loop, and some code after the loop has ended to process the results. To be clear, we **ARE NOT** using k-fold cross validation.

For each of the following steps in the cross-validation process, identify (by writing in the blank space) whether it occurs:

(A) prior to the for loop

(B) within the for loop

(C) after the for loop

(D) never in the CV process

C Average RMSE across all test-training splits and report

D Average the predictions and record their value

D Choose the model with the largest RMSE

D Choose the model with the smallest RMSE

B Compare the predictions from each model (that was fitted to training data) to the observed test data and compute RMSE

D Compute each model's AIC and BIC

D Fit each model using the testing data

A Fit each model using the training data

D Initialize vectors for each model that you will use to store their RMSEs

D Refit each model using the full data set and compare the coefficient estimates to the true values

A Split the data into test and training sets

B Use the models fitted to the training data to get predictions for the test data

END OF EXAM

Scratch Paper

12.

Confidence Interval: $0.007417 \pm 1.955(0.001871)$

$+ : 0.011074805$

$- : 0.003759195$

$(0.003759195, 0.011074805)$

$100(e^{0.472892} - 1)$

18.

$n - p - 1$

$n = 58$

$p = 2$

$58 - 2 - 1 = 55$ df

Scratch Paper

STAT 330-001

40142034 62

Barrett, Hayden Andrew

535731335 hbarret1 559-872-8876

Test: 641485 Form: A Wadsworth, Spencer Gordon

Entry: 07:23pm 10/21/2026 Checked out:

Employee: Julia L.

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NS

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062

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(D) Ridge regression often handles multicollinearity well

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(F) They can take advantage of the bias-variance tradeoff to produce more stable and accurate predictions

Question 8: Why are LASSO and ridge regression sometimes referred to as "shrinkage" methods? Specifically, what is being shrunk and how is this accomplished?

Unnhh... pass ~~Coefficients~~

JK everyone knows that it's referred to as the shrinkage method because β is getting shrunk down or at least something else is shrinking that causes β to shrink down. How does this happen?? Great question it is accomplished because of the penalizer which is β^2 for Ridge and $|\beta|$ for lasso.

$\frac{4}{100} = .04$

Question 9: A multiple linear regression model with $p = 3$ predictors is fit using $n = 100$ observations. Observation i has standardized residual $r_i = 2.4$ and leverage $h_{ii} = 0.25$.

Use it gives it to us... $D_i = \frac{r_i^2}{p+1}\left(\frac{h_{ii}}{1-h_{ii}}\right)$. $\frac{(2.4)^2}{4} \cdot \frac{.25}{1-.25}$ $\frac{5.76}{4} \cdot .33 \approx .48$ 1.44

(Circle one answer) Which conclusion is most appropriate?

(A) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

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$\hat{\beta} = (X^T X)^{-1} X^T y$

(B) What is the conceptual difference between $\widehat{\beta}$ and β?

~~one has a hat and the other one doesn't~~ I hope you don't mind jokes while grading (B)

$\hat{\beta}$ is the coefficient for each of our predictors ie how much of a change does a 1 unit increase in x_i have on our response for this sample. Key word here is sample. β on the other hand is similar to $\hat{\beta}$ except β is for the whole population, Because of this β is generally unknown. So β is the true coefficient for our predictors

(C) Explain how least squares can be used to estimate regression coefficients. You do not need to derive anything, but you should be clear about how this is done conceptually.

SSE $(y - X\hat{\beta})^T (y - X\hat{\beta}) \rightarrow (y^T - x^T \hat{\beta}^T)(y - X\hat{\beta})$ in this case $X^T = X$, $\hat{\beta}^T = \hat{\beta}$, $y^T = y$

deriv $y^T y - 2\hat{\beta}^T X^T y + \hat{\beta}^T X^T X \hat{\beta}$

$-2X^T y + 2\hat{\beta} X^T X$

$(X^T X)^{-1} 2\hat{\beta} X^T X = 2X^T y (X^T X)^{-1}$

$\hat{\beta} = X^T y (X^T X)^{-1}$

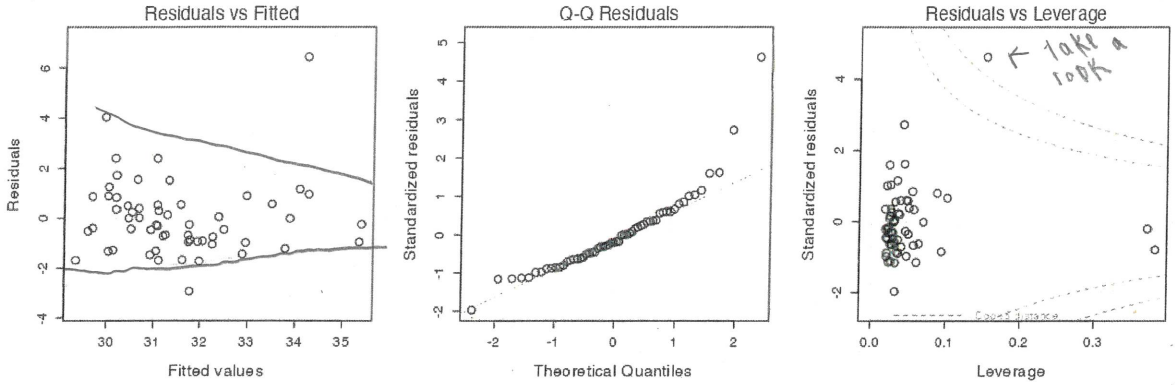
Thought it would be easier just to write it out

PART II

Some consider competitive eaters to be among the most inspiring individuals in modern times. We are interested in knowing what factors relate to a competitive eater's ability to consume. In a recent state fair, a contest was held where contestants competed to see who could eat the most hotdogs in 10 minutes. Data were collected on 58 amateur competitive eaters, most of whom were from the 4 largest counties in the state. For each eater, the total number of hotdogs consumed in a single 10 minute sitting was recorded. Each contestant's height in inches was also recorded, and each contestant was asked "How many minutes per week do you spend training for a contest?" The variables are `hotdogs`, `height`, and `minutes`. Model (1) is used for analysis. Use the plots and model fit output below to answer questions 11-13.

Model 1

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \texttt{minutes}_i + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{1}$$



Coefficient	Estimate	Std. Error	t value	Pr($>\|t\|$)
Intercept	-1.295392	5.467711	-0.237	0.813602
`height`	0.472892	0.080352	5.885	2.46e-07
`minutes`	0.007417	0.001871	3.963	0.000215

Other Summaries	Value
Breusch-Pagan p-value	0.071
Shapiro-Wilk p-value	2.43e-8
CV error	1.84623

(or might)

Question 11: State the estimated value of β_1, and interpret its value in the context of the problem

estimated value of $\beta_1 = .47$; holding all other predictors constant a 1 inch increase in X_1^V is associated with another .47 hot dogs consumed in the 10 minutes sitting.

$.007 \pm 3.96 \cdot .001$

Question 12: **(Circle all that apply)** Which of the following are true?

(A) The 95% confidence interval of β_2 does **not** contain 0

(B) The 95% confidence interval of β_2 does contain 0

(C) There is an 81.4% probability that $\beta_0 = 0$

(D) $\beta_1 = 0.4729$

no we have estimated

Question 13: Given what you know, provide a brief assessment of how well the data satisfy each of the following multiple linear regression assumptions:

(A) The relationships between the explanatory and response variables are linear

- Definitely seeing some violations in the R VS F and AV Plots with some outliers, but no extremely strong U shape in the graphs which is good for the assumption to not be violated
- looking iffy

(B) There is independence between each observation

- One flag here is that they are all mostly sampled from the 4 largest counties in the state which is not good for ind. We want to avoid clustering in our sampling for this assumption not to be violatated
- looking iffy

(C) The errors are normally distributed

The Shapiro Wilk test p-value is very low supporting a violation along with the Q-Q plot having curves and especially on the upper end is supporting that this assumption is doing violatated

- looking pretty bad

(D) The errors have constant variance (homoscedasticity)

Our R vs F Plot is showing small signs of funneling, but nothing major. Our breusch-Pagan p-value is above .05 indicating that we have a constant error variance. Overall would say this assumption is mostly good, I wouldn't worry abt it

- looking good

(E) There are no unduly influential points

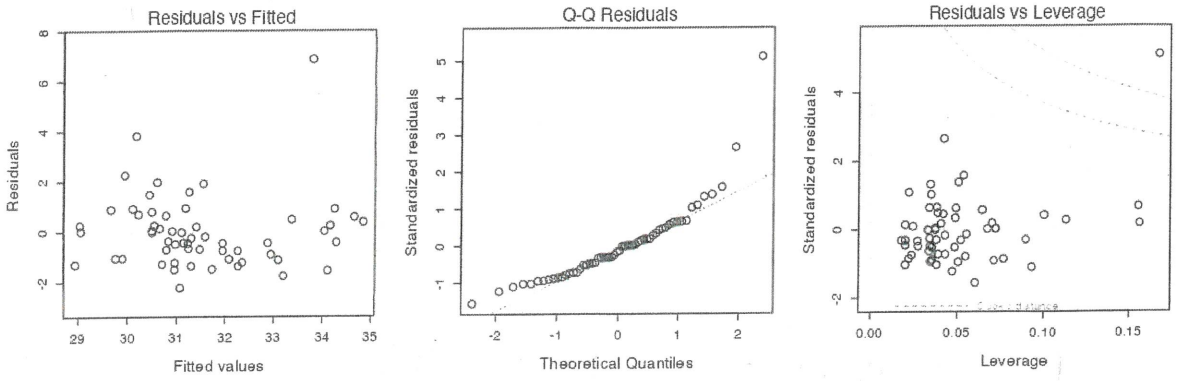
Given our Residual VS Leverage graph it would seem we have a point that is highly influential given it is past the cooks distance with a high leverage, high residual. While this has been slightly violated we should check how the data was collected and make sure the data is accurate before we throw the point out

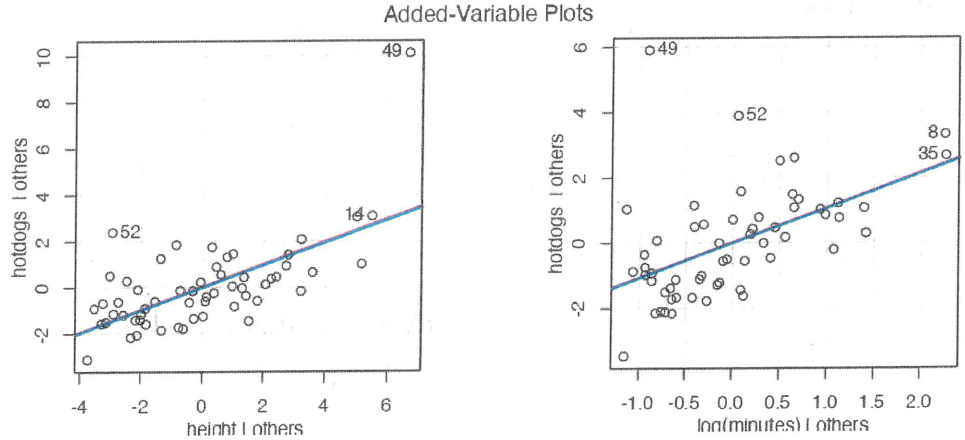
- looking iffy

Model 2

Model (2) below represents another possible multiple linear regression model for the same data. Here the variable `minutes` was log-transformed prior to fitting the model. Use model (2) and the output below to answer questions 14-16

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \log(\texttt{minutes}_i) + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{2}$$





Coefficient	Estimate	Std. Error	t value	$\Pr(>\|t\|)$
Intercept	-6.05589	5.43620	-1.114	0.27
`height`	0.48834	0.07821	6.244	6.48e-08
log(`minutes`)	1.06702	0.24079	4.431	4.51e-05

Other Summaries	Value
Breusch-Pagan p-value	0.028
Shapiro-Wilk p-value	3.43e-5
CV RMSE	1.66702

Question 14: A Breusch-Pagan test for model (1) results in a p-value of 0.071 while a Breusch-Pagan test for model (2) results in a p-value of 0.028. A researcher concludes that model (1) is clearly superior to model (2). How would you respond?

Well just from 1 test I would not conclude that one model is better than the other, while it is true that theoretically a higher Breusch Pagan P-Value and it being above .05 is better we still have all the other LINE assumptions to check, Along with that we don't have all of our tools at disposal for LINE assumptions to go in more depth

Question 15: Explain what advantage(s) and disadvantage(s) model (2) provides over model (1)

Well theoretically transforming it should help with linearity and we do see the av plot for log(minutes) look a lot better. It should also help stabilize error variance and lower right skew in residuals. Though in Comparison model 1 definitely fits the constant error variance (E) assumption better, The interpretation for model 2 will also be different now

Question 16: State the estimated value of β_2 and provide an interpretation in context

$\beta_2 = \frac{1.067}{100} = .01067$

Keeping all other predictors constant a 1% increase in minutes per week training for this contest is associated with a .01067 increase in hot dogs consumed during the 10 minute sitting.

Use the following for questions 17-19

Your best friend reveals that he will be competing in the hotdog eating contest at the next state fair. You know that he is 68 inches tall and that he trains 79 minutes per week. Note that log(79) = 4.369. Use this information and the fit for model (2) to answer the following questions.

Question 17: Use model (2) to calculate the predicted number of hotdogs your friend can consume in 10 minutes

$y_1 = -6.06 + 68 \cdot .49 + 1.07 \cdot 4.37$

$= -6.06 + 33.32 + 4.68$

$= \boxed{31.94 \text{ hot dogs}}$

Question 18: To create a confidence or prediction interval for the expected number of hotdogs your friend can consume in 10 minutes, what is the correct number of degrees of freedom for the critical t^* used to calculate the margin of error? $58 - 2 - 1 = 55$ I assume we are not adding him and $n = 58$ still

(A) 55

(B) 57

(C) 56

(D) 48

Question 19: You're given that $s\sqrt{x^*(X^\top X)^{-1}x^*} = 0.204$, $s\sqrt{1 + x^*(X^\top X)^{-1}x^*} = 1.485$, and $t^*_{df,0.975} = 2.004$. Give a 95% *prediction* interval for the number of hotdogs your friend can consume in 10 minutes.

$31.94 \pm 2.004 \cdot 1.485$

31.94 ± 2.98

$28.96, 34.92$

~~We would predict that 95% of the time~~

We are 95% sure that the true amount of hot dogs that my friend will consume in the 10 minute sitting at the hot dog eating contest at the next State fair is between 28.96 and 34.92 hot dogs

Part III - Application

Questions 20-24 concern the following dataset:

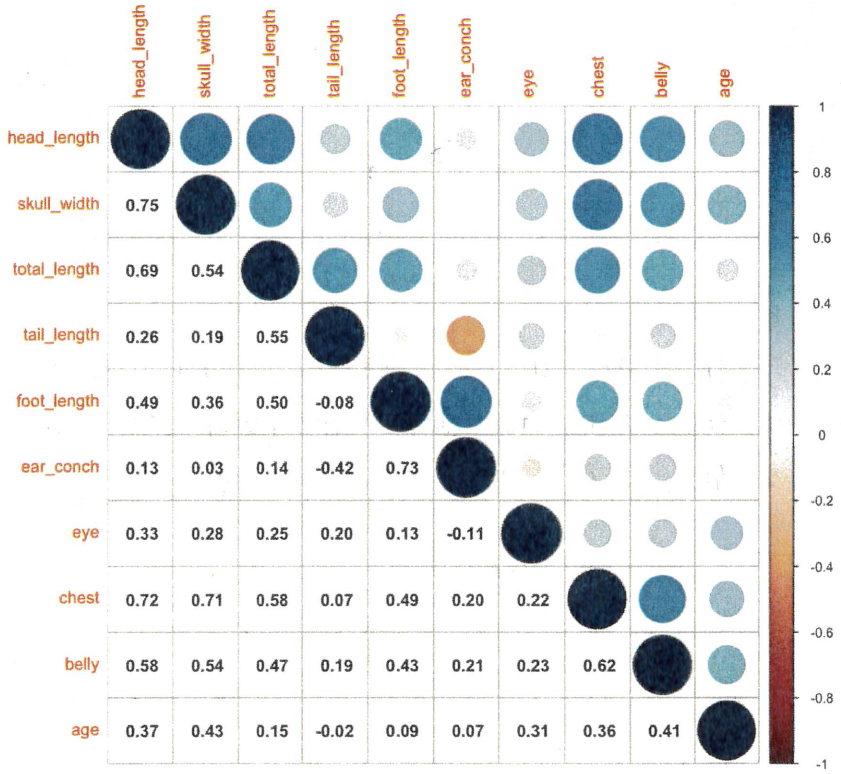
In 1995, Australian zoologists conducted a study of local possum populations. They trapped dozens of specimens, determined their ages, and took measurements of several physical attributes from each possum. The process of determining a wild animal's age can be challenging and imprecise, and usually requires a detailed analysis of the animal's teeth, skin, skeletal structure, or DNA. Among other questions, researchers were interested in whether a possum's age could be predicted using only easy to obtain physical measurements. For this analysis we will focus on the 59 male possums from the sample. The variables in the dataset are as follows (see below). A multiple linear regression model to predict age using all other variables was fit to these data, resulting in the following model summary and coefficient estimates (see below)

Variable	Description
head_length	in mm
skull_width	in mm
total_length	in cm
tail_length	in cm
foot_length	in mm
ear_conch	ear conch circumference (in mm)
eye	width of right eye (in mm)
chest	chest girth (in cm)
belly	belly girth (in cm)
age	in years

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -13.22377    7.34105  -1.801   0.0778 .
head_length   0.07077    0.10120   0.699   0.4877
skull_width   0.16559    0.10599   1.562   0.1247
total_length -0.04752    0.10167  -0.467   0.6423
tail_length  -0.05049    0.17030  -0.296   0.7681
foot_length  -0.21499    0.10733  -2.003   0.0507 .
ear_conch     0.15730    0.10231   1.538   0.1306
eye           0.43736    0.22549   1.940   0.0582 .
chest         0.01883    0.18154   0.104   0.9178
belly         0.20983    0.11580   1.812   0.0761 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.655 on 49 degrees of freedom
Multiple R-squared:  0.3538,    Adjusted R-squared:  0.2352
F-statistic: 2.982 on 9 and 49 DF,  p-value: 0.006552
```

A correlation plot (shown below) of these variables points towards some level of multicollinearity in the data. The maximum VIF among these variables is approximately 4 with an average VIF of approximately 3.



Question 20:

(a) Using the information from the summary table above, produce a 95% confidence interval for the coefficient on `skull_width`. The 0.975 critical value of a t-distribution with 49 degrees of freedom is 2.01. You may report your answer in the form $a \pm b * c$.

$.166 \pm 2.01 \cdot .106$

(b) A 95% confidence interval for $\beta_{\texttt{belly}}$, the coefficient on belly width, is (-0.023, 0.443). Interpret this interval within the context of the data.

Don't know which one you want so here you go, if I must choose I choose 1

Since 0 is inside the 95% C.I. we fail to reject that $\beta_{belly} = 0$ and conclude that belly girth in cm has a significant effect on a possum's age after accounting for other predictors

holding all other predictors constant we are 95% confident that a 1 cm increase in belly girth is associated with a change between (-.023, .443) for the possums age.

Question 21: Note from the model summary that while the overall F-test of model significance produces a p-value of 0.0065, none of the individual model coefficients are found to have a significant effect at the $\alpha = 0.05$ level. What do you think is driving this "contradiction"? (It's not actually a contradiction.)

When the Indpence is violated we can see deflated p-values, since we see a large amount of multicollinearity that is directly related to a lack of independence,

Question 22: Briefly describe how the "best subsets" algorithm using AIC as its criterion performs variable selection.

Well the best subset algothrim is either going to start with just β_0 or your full model (whatever is better in specific case) and it well than either add or take away a predictor continuously until it produces the best model. AIC is going to come in and help fix the model using 2 · # of predictors as it's limiter (don't remember the word)

lower adj R^2

Question 23: Best subsets using AIC as a criterion identified the following model with five variables:
`lm(age ~ skull_width + foot_length + ear_conch + eye + belly, data = possum)`
This model produces the following summary and coefficient estimates:

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -14.76661    4.97828  -2.966  0.00451 **
skull_width   0.20221    0.07990   2.531  0.01439 *
foot_length  -0.22936    0.09385  -2.444  0.01789 *
ear_conch     0.18239    0.08653   2.108  0.03979 *
eye           0.45001    0.21687   2.075  0.04285 *
belly         0.21405    0.10316   2.075  0.04287 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.609 on 53 degrees of freedom
Multiple R-squared:  0.3396,    Adjusted R-squared:  0.2773
F-statistic:  5.45 on 5 and 53 DF,  p-value: 0.0004056
```

Is this model guaranteed to have the highest adjusted-R^2 among all possible models.

(A) TRUE

BIC is ↑ when $n > 8$

(B) FALSE

Question 24: Suppose we want to compare the out-of-sample predictive performance of the full model and the model chosen by best subsets, and ultimately select the model with the smaller cross-validated RMSE.

Cross-validation using repeated random training-test splits can be structured around a for loop over some number of iterations (let's say 100), with some setup happening prior to the start of the loop, the bulk of the code within the loop, and some code after the loop has ended to process the results. To be clear, we **ARE NOT** using k-fold cross validation.

For each of the following steps in the cross-validation process, identify (by writing in the blank space) whether it occurs:

(A) prior to the for loop

(B) within the for loop

(C) after the for loop

(D) never in the CV process

C Average RMSE across all test-training splits and report

B Average the predictions and record their value

D Choose the model with the largest RMSE

C Choose the model with the smallest RMSE

C Compare the predictions from each model (that was fitted to training data) to the observed test data and compute RMSE

D Compute each model's AIC and BIC

B Fit each model using the testing data

B Fit each model using the training data

A Initialize vectors for each model that you will use to store their RMSEs

C Refit each model using the full data set and compare the coefficient estimates to the true values

A Split the data into test and training sets

B Use the models fitted to the training data to get predictions for the test data

END OF EXAM

STAT 330-001

40142034 57

Curtis, Logan Jimmy

388631034 ljc337 407-401-2352

Test: 641362 Form: A Wadsworth, Spencer Gordon

Entry: 06:51pm 10/21/2026 Checked out:

Employee: Braxton S.

SPECIAL INSTRUCTIONS

TC Calc
Notes: 3x5
Dictionary
Take a deep breath. Be careful to read instructions thoroughly. Answer questions completely.

Please keep this sheet **VISIBLE** on your desk at all times. When you are finished, *staple this sheet* to the front of your test booklet and return all materials to the Grading Table. You may use the space below for notes or scratch paper.

PLEASE TURN OFF YOUR CELL PHONE

$\frac{(2.4)^2}{3+1}\left(\frac{0.25}{1-0.25}\right) = \frac{5.76}{4}\left(\frac{0.25}{0.75}\right) = 1.44(0.\overline{33}) = 0.48$

$\frac{4}{n} = \frac{4}{100} = 0.04$

CI of B_2 → $\pm t^* \times SE$

$t = \frac{\hat{B}_j - B_{j,0}}{SE(\hat{B}_j)}$

$2.43e^{-8}$

+c% in x_i gives $B_i \log(1+c/100)$ units in mean y_i

+1.06702% in minutes gives $B_2 \log(1.0106702)$ hotdogs in mean hotdogs;

95% CI for skull-width

1.06702

2.004 ± 1.485

$X: n \times (p+1)$

$t = \frac{\hat{B}_j - B_{j,0}}{SE(\hat{B}_j)}$

$s\sqrt{1 + x^*(x^Tx)^{-1}x^*}$

$s = \sqrt{\frac{SSE}{49}}$ $SSE = \sum e_i^2$ $s = 0.08377742$

$\hat{B} = 0.16559$

$SE = 0.10599$

$\frac{0.16559 - (-13.22377)}{0.10599}$

they gave it to me bruh

$2.01 \pm 6^*c$ $s\sqrt{1 + x^{*T}(x^Tx)^{-1}x^*}$

$n - p - 1 = 49$ $X: 59 \times 10$

$59 - 9 - 1 = 49$ $X^TX: 10 \times 10$

n p

0.005008393/
0.027420048/
0.00225815/
0.00254924/
0.04622071/
0.02479329/
0.19128377/
0.00035456/
0.04402862/

0.3439167 89

NS
40142034
057

STAT 330 - Exam 1

Section 001 (Wadsworth), Section 002 (White)
October 16th - 21st, 2026

Instructions

- You may not refer to your notes, other people, or other resources. You can, however, use one 3 inch by 5 inch note card that you create (writing on both sides is allowed).
- Do not discuss any aspect of this exam with anyone except a Stat 330 TA or instructor until scores are posted.
- You are allowed to use a calculator for this exam.
- In Parts II and III, **make sure to clearly show your work for any calculations that you make and to give your answers in context of the data sets.** Place your final answer inside the box, if one is provided.
- There are 24 questions. Make sure you answer all of them.
- There is extra scratch paper at the end of the exam.

PART I

Question 1: (Circle one answer) Partial regression (or added-variable) plots illustrate the effect of adding an additional variable to the model after accounting for the variables that are already included in the model.

(A) TRUE

(B) FALSE

Question 2: (TRUE/FALSE) If the relationship between x and y is nonlinear, a simple linear regression model may still be feasible after transforming the data.

- TRUE
- FALSE

Question 3: (Circle one answer) When adding an additional variable to a linear regression model, the model's R^2 ...

(A) always increases or remains unchanged.

(B) always decreases or remains unchanged.

(C) may increase or decrease depending of the additional variable.

Question 4: (Circle all that apply) Which of the following statements describe possible effects of multicollinearity?

(A) It can cause the linearity assumption to be violated.

(B) It can cause estimated coefficients to have signs opposite those of their true population coefficients.

(C) It can result in larger standard errors, and thus greater uncertainty in estimates.

(D) It can cause the coefficient of determination (R^2) to become negative.

Question 5: (Circle all that apply) Which of the following are true statements about multiple linear regression (MLR)?

(A) BIC will usually favor a model with more predictors than the AIC criterion when performing model selection.

(B) Cook's distance can be used to identify influential observations.

(C) Transformations of the x's or y may help alleviate problems with linearity, equal variance, or normality.

(D) The least-squares estimate of β can be obtained by fitting a simple linear regression for each x_k only.

(E) The assumptions on the error variables (i.e., the ϵ_i) in a multiple linear regression model are exactly the same as they are for simple linear regression.

Question 6: (Circle one answer) What is the primary benefit of using cross-validation (CV)?

(A) CV reduces variance in the sampling distribution of regression coefficients by splitting the training and test data.

(B) CV gives a more realistic representation of model performance by predicting observations that were not used to fit the model.

(C) CV ensures model assumptions are met which eliminates a time-consuming step of the modeling process.

(D) CV is the most efficient way to calculate the shrunken coefficient estimates in LASSO and ridge regression models.

Question 7: (Circle all that apply) In datasets with high multicollinearity and/or redundant covariates, what advantages do LASSO and/or ridge regression have in comparison to the traditional ordinary least squares (OLS) approach to model fitting?

(A) LASSO can set coefficients to be exactly equal to zero, and thus serves as an automated approach to variable selection

(B) Their coefficient estimates are less biased than those of OLS

(C) They produce more accurate confidence intervals and hypothesis tests

(D) Ridge regression often handles multicollinearity well

(E) They can handle datasets where the number of predictors exceeds the number of observations

(F) They can take advantage of the bias-variance tradeoff to produce more stable and accurate predictions

Question 8: Why are LASSO and ridge regression sometimes referred to as "shrinkage" methods? Specifically, what is being shrunk and how is this accomplished?

LASSO and Ridge regression are known as "shrinkage" methods because they shrink the coefficients of the predictors towards 0 (with LASSO they can reach 0). This is accomplished first by finding Lambda (λ) using cross-validation. Then we add all the squared differences of the response i's and the beta nulls with the beta transpose x i's. Then we add lambda times the summation of either the Beta j's squared (for ridge) or the absolute values of the Beta j's (For LASSO).

Question 9: A multiple linear regression model with $p = 3$ predictors is fit using $n = 100$ observations. Observation i has standardized residual $r_i = 2.4$ and leverage $h_{ii} = 0.25$.

Use

$$D_i = \frac{r_i^2}{p+1}\left(\frac{h_{ii}}{1-h_{ii}}\right).$$

(Circle one answer) Which conclusion is most appropriate?

(A) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(B) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(C) $D_i = 0.48$. The observation is not influential because its leverage is less than 0.50.

(D) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(E) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(F) $D_i = 0.2$. The observation is not influential because its leverage is less than 0.50.

(G) Cook's distance cannot be calculated without knowing the observation's original residual.

Question 10:

(A) What is the formula for $\widehat{\beta}$?

$\hat{\beta} = (X^T X)^{-1} X^T Y$

(B) What is the conceptual difference between $\widehat{\beta}$ and β?

β is the model parameter you want to estimate while $\hat{\beta}$ is the result of estimation. $\hat{\beta}$, more specifically is an estimated parameter that comes from the data.

(C) Explain how least squares can be used to estimate regression coefficients. You do not need to derive anything, but you should be clear about how this is done conceptually.

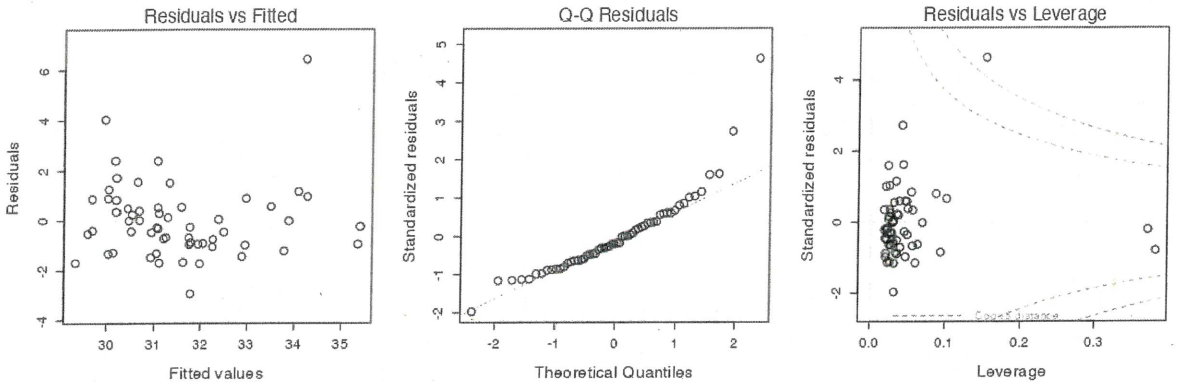
Least squares finds the effects the predictors have on the response variable and gives coefficients equal to what a single unit increase of that specific predictor would change in the response.

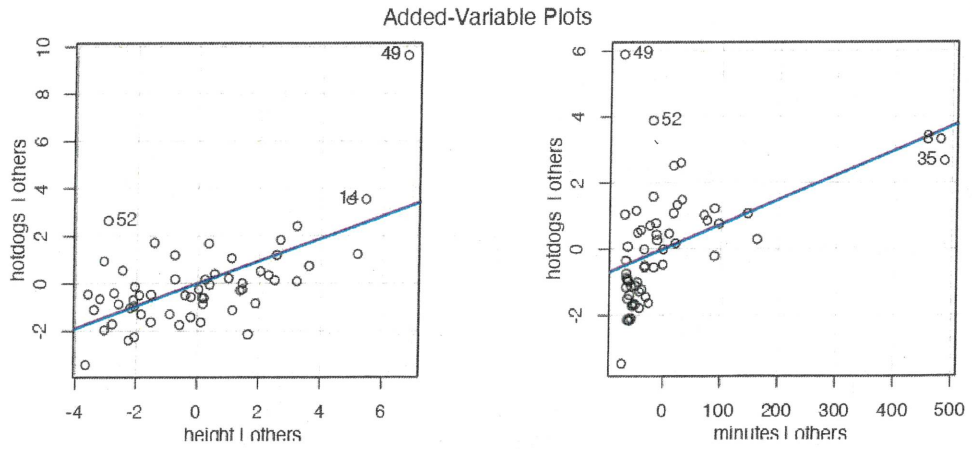
PART II

Some consider competitive eaters to be among the most inspiring individuals in modern times. We are interested in knowing what factors relate to a competitive eater's ability to consume. In a recent state fair, a contest was held where contestants competed to see who could eat the most hotdogs in 10 minutes. Data were collected on 58 amateur competitive eaters, most of whom were from the 4 largest counties in the state. For each eater, the total number of hotdogs consumed in a single 10 minute sitting was recorded. Each contestant's height in inches was also recorded, and each contestant was asked "How many minutes per week do you spend training for a contest?" The variables are `hotdogs`, `height`, and `minutes`. Model (1) is used for analysis. Use the plots and model fit output below to answer questions 11-13.

Model 1

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \texttt{minutes}_i + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{1}$$





Coefficient	Estimate	Std. Error	t value	$\Pr(>\|t\|)$
Intercept	-1.295392	5.467711	-0.237	0.813602
`height`	0.472892	0.080352	5.885	2.46e-07
`minutes`	0.007417	0.001871	3.963	0.000215

Other Summaries	Value
Breusch-Pagan p-value	0.071
Shapiro-Wilk p-value	2.43e-8
CV error	1.84623

Question 11: State the estimated value of β_1, and interpret its value in the context of the problem

Estimated $\beta_1 = 0.472892$

Holding all other predictors constant, a one-inch increase in height should cause a 0.472892 increase in the amount of hotdogs eaten in a 10-minute period.

Question 12: **(Circle all that apply)** Which of the following are true?

(A) The 95% confidence interval of β_2 does **not** contain 0

(B) The 95% confidence interval of β_2 does contain 0

(C) There is an 81.4% probability that $\beta_0 = 0$

(D) $\beta_1 = 0.4729$

Question 13: Given what you know, provide a brief assessment of how well the data satisfy each of the following multiple linear regression assumptions:

(A) The relationships between the explanatory and response variables are linear

The relationships between the explanatory and response variables are a bit iffy on linearity. On the residuals vs fitted graph, there is a general downward slope, but the points are still somewhat spread across in a linear fashion. However, the add-variables plots show general upward trends that point out issues in linearity.

(B) There is independence between each observation

The data was collected on contestants from the same contest and the same state but it seems to have been collected in no specific order. So as long as each contestant was asked in a random order, the data should have good independence between observations

(C) The errors are normally distributed

The Q-Q plot shown suggests normality due to most points following the same general slope. However, the p-value of the Shapiro-Wilk test is significantly small at about 2.43e-8, which suggests there could be an issue with normality worth investigating.

(D) The errors have constant variance (homoscedasticity)

The Breusch-Pagan test produced a p-value of 0.071 which suggests general homoscedasticity, however the added-variable plots shows something different. For the height av-plot the points look equally spread, but for the minutes av-plot there is a huge collection of points on the left, which suggests unconstant variance in regards to the minutes predictor.

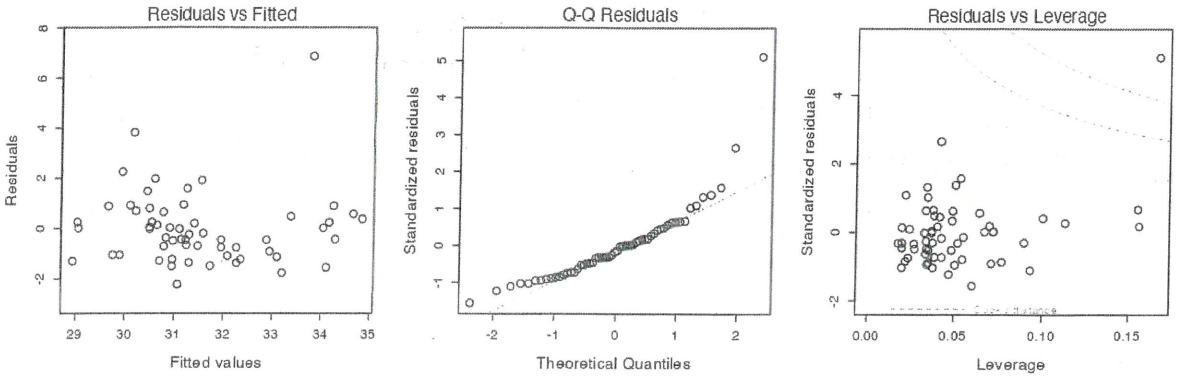
(E) There are no unduly influential points

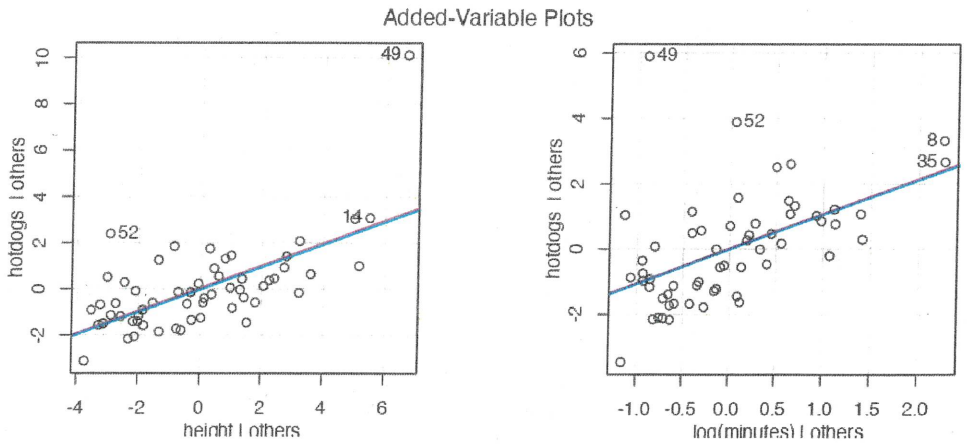
While there are two observations with higher leverage than the rest of the observations, I don't think that those two points are unduly influential because the rest of the points don't follow those two points much.

Model 2

Model (2) below represents another possible multiple linear regression model for the same data. Here the variable `minutes` was log-transformed prior to fitting the model. Use model (2) and the output below to answer questions 14-16

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \log(\texttt{minutes}_i) + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{2}$$





Coefficient	Estimate	Std. Error	t value	$\Pr(>\|t\|)$
Intercept	-6.05589	5.43620	-1.114	0.27
`height`	0.48834	0.07821	6.244	6.48e-08
log(`minutes`)	1.06702	0.24079	4.431	4.51e-05

Other Summaries	Value
Breusch-Pagan p-value	0.028
Shapiro-Wilk p-value	3.43e-5
CV RMSE	1.66702

Question 14: A Breusch-Pagan test for model (1) results in a p-value of 0.071 while a Breusch-Pagan test for model (2) results in a p-value of 0.028. A researcher concludes that model (1) is clearly superior to model (2). How would you respond?

The results of the Breusch-Pagan test can really only suggest that there is an issue with equal variance, and after checking the added-variable plots, model(2)'s equal variance looks more sound than model(1)'s because (2) fixes (1)'s issue with the minutes av-plot. Even then, we still need to take into account all the other advantages/disadvantages of the two models and choose which model will better help us achieve our goal more accordingly.

Question 15: Explain what advantage(s) and disadvantage(s) model (2) provides over model (1)

While equal variance might still have issues in model(2), the homoscedasticity of model(2) looks much more sound than model(1). On the other hand, model (2) has a bit more leverage in each observation which suggests there's more likely an effect by influential points, even if the points with the highest leverage are lower. On top of that, normality looks ever-so-slightly less secure in model(2), but it shouldn't be an issue.

Question 16: State the estimated value of β_2 and provide an interpretation in context

Estimated $\beta_2 = 1.06702$ $\quad \log(1 + {}^{1.06702}/_{100}) = \log(1.0106702)$

A 1.06702% increase in minutes adds $\log(1.0106702)$ hotdogs to the mean value of the hotdogs.

Use the following for questions 17-19

Your best friend reveals that he will be competing in the hotdog eating contest at the next state fair. You know that he is 68 inches tall and that he trains 79 minutes per week. Note that log(79) = 4.369. Use this information and the fit for model (2) to answer the following questions.

Question 17: Use model (2) to calculate the predicted number of hotdogs your friend can consume in 10 minutes

$\text{hotdogs}_i = \hat{\beta}_0 + \hat{\beta}_1 \text{height}_i + \hat{\beta}_2 \log(\text{minutes}_i)$ $\quad e_i = Y_i - \hat{Y}_i$

$\hat{\beta}_0 + \hat{\beta}_1 68 + \hat{\beta}_2 \log(79)$

$-6.05589 + 0.48834(68) + 1.06702(4.369)$

$-6.05589 + 33.20712 + 4.66181038$

$= 31.81304038$

My friend will most likely be able to eat 31 whole hotdogs.

Question 18: To create a confidence or prediction interval for the expected number of hotdogs your friend can consume in 10 minutes, what is the correct number of degrees of freedom for the critical t^* used to calculate the margin of error?

(A) 55 $\quad df = n - p - 1$

(B) 57 $\quad 55 = 58 - 2 - 1$

(C) 56

(D) 48

Question 19: You're given that $s\sqrt{x^*(X^\top X)^{-1}x^*} = 0.204$, $s\sqrt{1 + x^*(X^\top X)^{-1}x^*} = 1.485$, and $t^*_{df,0.975} = 2.004$. Give a 95% *prediction* interval for the number of hotdogs your friend can consume in 10 minutes.

Use ± 1.485 for prediction

$(2.004 - 1.485, \; 2.004 + 1.485)$

$(0.519, \; 3.489)$

95% CI doesn't include 0, so p-value is less than 0.05, so there might be an issue with this prediction.

There is a 95% chance that our prediction of the amount of hotdogs my friend can eat in 10 minutes is within between +0.519 and +3.489 of 31.81304038 hotdogs.

Part III - Application

Questions 20-24 concern the following dataset:

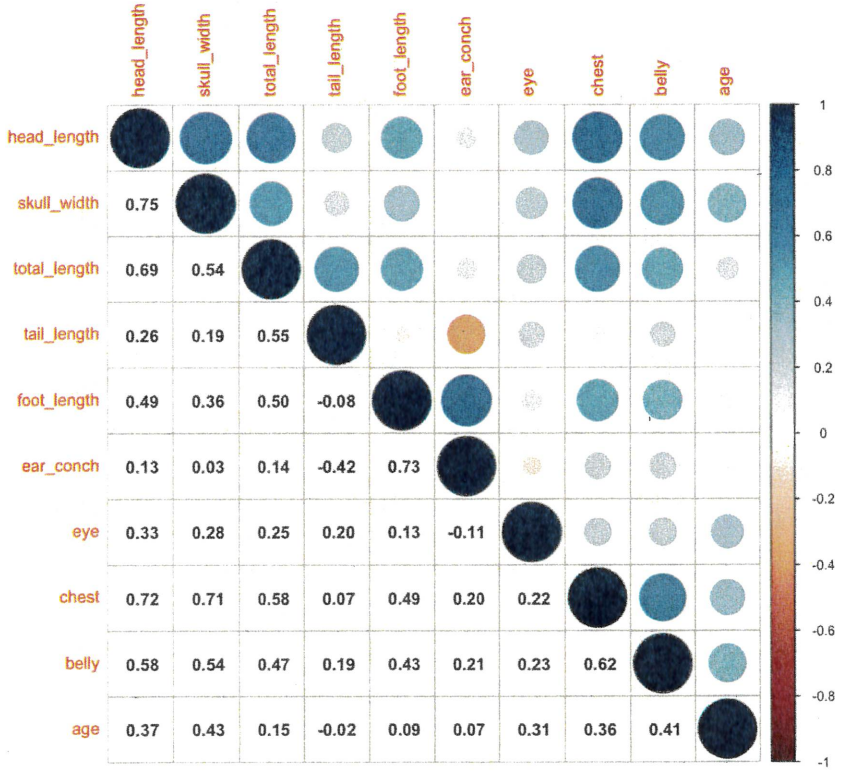
In 1995, Australian zoologists conducted a study of local possum populations. They trapped dozens of specimens, determined their ages, and took measurements of several physical attributes from each possum. The process of determining a wild animal's age can be challenging and imprecise, and usually requires a detailed analysis of the animal's teeth, skin, skeletal structure, or DNA. Among other questions, researchers were interested in whether a possum's age could be predicted using only easy to obtain physical measurements. For this analysis we will focus on the 59 male possums from the sample. The variables in the dataset are as follows (see below). A multiple linear regression model to predict age using all other variables was fit to these data, resulting in the following model summary and coefficient estimates (see below)

Variable	Description
head_length	in mm
skull_width	in mm
total_length	in cm
tail_length	in cm
foot_length	in mm
ear_conch	ear conch circumference (in mm)
eye	width of right eye (in mm)
chest	chest girth (in cm)
belly	belly girth (in cm)
age	in years

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  -13.22377    7.34105  -1.801   0.0778 .
head_length    0.07077    0.10120   0.699   0.4877
skull_width    0.16559    0.10599   1.562   0.1247
total_length  -0.04752    0.10167  -0.467   0.6423
tail_length   -0.05049    0.17030  -0.296   0.7681
foot_length   -0.21499    0.10733  -2.003   0.0507 .
ear_conch      0.15730    0.10231   1.538   0.1306
eye            0.43736    0.22549   1.940   0.0582 .
chest          0.01883    0.18154   0.104   0.9178
belly          0.20983    0.11580   1.812   0.0761 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.655 on 49 degrees of freedom
Multiple R-squared:  0.3538,    Adjusted R-squared:  0.2352
F-statistic: 2.982 on 9 and 49 DF,  p-value: 0.006552
```

A correlation plot (shown below) of these variables points towards some level of multicollinearity in the data. The maximum VIF among these variables is approximately 4 with an average VIF of approximately 3.



Question 20:

(a) Using the information from the summary table above, produce a 95% confidence interval for the coefficient on `skull_width`. The 0.975 critical value of a t-distribution with 49 degrees of freedom is 2.01. You may report your answer in the form $a \pm b * c$.

2.01 ± 1.655

(b) A 95% confidence interval for $\beta_{\texttt{belly}}$, the coefficient on belly width, is (-0.023, 0.443). Interpret this interval within the context of the data.

We are 95% confident that when predicting age using belly width, well will be within -0.023 and +0.443 of the age in years of a possum.

Question 21: Note from the model summary that while the overall F-test of model significance produces a p-value of 0.0065, none of the individual model coefficients are found to have a significant effect at the $\alpha = 0.05$ level. What do you think is driving this "contradiction"? (It's not actually a contradiction.)

I believe multicollinearity is what's causing the "contradiction" in the F-test and t-tests. Multicollinearity can make t-tests lose their power, and multicollinearity is already confirmed to be an issue due to the correlation plot.

Question 22: Briefly describe how the "best subsets" algorithm using AIC as its criterion performs variable selection.

Best-subsets fits every possible model using $n/\log(SSE/n)+2k$ and chooses models with most-likely the best out-of-sample prediction power. Uses in sample data and will not really worry about a larger amount of predictors.

Question 23: Best subsets using AIC as a criterion identified the following model with five variables:
`lm(age ~ skull_width + foot_length + ear_conch + eye + belly, data = possum)`
This model produces the following summary and coefficient estimates:

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -14.76661    4.97828  -2.966  0.00451 **
skull_width   0.20221    0.07990   2.531  0.01439 *
foot_length  -0.22936    0.09385  -2.444  0.01789 *
ear_conch     0.18239    0.08653   2.108  0.03979 *
eye           0.45001    0.21687   2.075  0.04285 *
belly         0.21405    0.10316   2.075  0.04287 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.609 on 53 degrees of freedom
Multiple R-squared:  0.3396,    Adjusted R-squared:  0.2773
F-statistic:  5.45 on 5 and 53 DF,  p-value: 0.0004056
```

Is this model guaranteed to have the highest adjusted-R^2 among all possible models.

(A) TRUE

(B) FALSE

Question 24: Suppose we want to compare the out-of-sample predictive performance of the full model and the model chosen by best subsets, and ultimately select the model with the smaller cross-validated RMSE.

Cross-validation using repeated random training-test splits can be structured around a for loop over some number of iterations (let's say 100), with some setup happening prior to the start of the loop, the bulk of the code within the loop, and some code after the loop has ended to process the results. To be clear, we **ARE NOT** using k-fold cross validation.

For each of the following steps in the cross-validation process, identify (by writing in the blank space) whether it occurs:

(A) prior to the for loop

(B) within the for loop

(C) after the for loop

(D) never in the CV process

__C__ Average RMSE across all test-training splits and report

__B__ Average the predictions and record their value

__D__ Choose the model with the largest RMSE

__C__ Choose the model with the smallest RMSE

__B__ Compare the predictions from each model (that was fitted to training data) to the observed test data and compute RMSE

__B__ Compute each model's AIC and BIC

__D__ Fit each model using the testing data

__B__ Fit each model using the training data

__A__ Initialize vectors for each model that you will use to store their RMSEs

__C__ Refit each model using the full data set and compare the coefficient estimates to the true values

__A__ Split the data into test and training sets

__C__ Use the models fitted to the training data to get predictions for the test data

END OF EXAM

STAT 330-001

40142034 72

Kirkham, Benjamin David

953189213 kirkham0 801-309-0570

Test: 641716 Form: A Wadsworth, Spencer Gordon

Entry: 08:38pm 10/21/2026 Checked out:

Employee: Addison R.

SPECIAL INSTRUCTIONS

TC Calc
Notes: 3x5
Dictionary
Take a deep breath. Be careful to read instructions thoroughly. Answer questions completely.

Please keep this sheet **VISIBLE** on your desk at all times. When you are finished, *staple this sheet* to the front of your test booklet and return all materials to the Grading Table. You may use the space below for notes or scratch paper.

PLEASE TURN OFF YOUR CELL PHONE

40142034
072

STAT 330 - Exam 1

Section 001 (Wadsworth), Section 002 (White)
October 16th - 21st, 2026

Instructions

- You may not refer to your notes, other people, or other resources. You can, however, use one 3 inch by 5 inch note card that you create (writing on both sides is allowed).
- Do not discuss any aspect of this exam with anyone except a Stat 330 TA or instructor until scores are posted.
- You are allowed to use a calculator for this exam.
- In Parts II and III, **make sure to clearly show your work for any calculations that you make and to give your answers in context of the data sets.** Place your final answer inside the box, if one is provided.
- There are 24 questions. Make sure you answer all of them.
- There is extra scratch paper at the end of the exam.

PART I

Question 1: (Circle one answer) Partial regression (or added-variable) plots illustrate the effect of adding an additional variable to the model after accounting for the variables that are already included in the model.

(A) TRUE

(B) FALSE

Question 2: (TRUE/FALSE) If the relationship between x and y is nonlinear, a simple linear regression model may still be feasible after transforming the data.

- TRUE
- FALSE

Question 3: (Circle one answer) When adding an additional variable to a linear regression model, the model's R^2 ...

(A) always increases or remains unchanged.

(B) always decreases or remains unchanged.

(C) may increase or decrease depending of the additional variable.

Question 4: (Circle all that apply) Which of the following statements describe possible effects of multicollinearity?

(A) It can cause the linearity assumption to be violated.

(B) It can cause estimated coefficients to have signs opposite those of their true population coefficients.

(C) It can result in larger standard errors, and thus greater uncertainty in estimates.

(D) It can cause the coefficient of determination (R^2) to become negative.

Question 5: (Circle all that apply) Which of the following are true statements about multiple linear regression (MLR)?

(A) BIC will usually favor a model with more predictors than the AIC criterion when performing model selection.

(B) Cook's distance can be used to identify influential observations.

(C) Transformations of the x's or y may help alleviate problems with linearity, equal variance, or normality.

(D) The least-squares estimate of β can be obtained by fitting a simple linear regression for each x_k only.

(E) The assumptions on the error variables (i.e., the ϵ_i) in a multiple linear regression model are exactly the same as they are for simple linear regression.

Question 6: (Circle one answer) What is the primary benefit of using cross-validation (CV)?

(A) CV reduces variance in the sampling distribution of regression coefficients by splitting the training and test data.

(B) CV gives a more realistic representation of model performance by predicting observations that were not used to fit the model.

(C) CV ensures model assumptions are met which eliminates a time-consuming step of the modeling process.

(D) CV is the most efficient way to calculate the shrunken coefficient estimates in LASSO and ridge regression models.

Question 7: (Circle all that apply) In datasets with high multicollinearity and/or redundant covariates, what advantages do LASSO and/or ridge regression have in comparison to the traditional ordinary least squares (OLS) approach to model fitting?

(A) LASSO can set coefficients to be exactly equal to zero, and thus serves as an automated approach to variable selection

(B) Their coefficient estimates are less biased than those of OLS

(C) They produce more accurate confidence intervals and hypothesis tests

(D) Ridge regression often handles multicollinearity well

(E) They can handle datasets where the number of predictors exceeds the number of observations

(F) They can take advantage of the bias-variance tradeoff to produce more stable and accurate predictions

Question 8: Why are LASSO and ridge regression sometimes referred to as "shrinkage" methods? Specifically, what is being shrunk and how is this accomplished?

They are refered to as shrinkage methods because they shrink down the values of the estimated $\hat{\beta}$'s. They can also reduce the size of the model by shrinking coefficients to 0, especially Lasso. It is accomplished by regressing the predictors on each other and then applying a penalty determined based on which method is chosen that shrinks estimated coefficents to reduce multicollinearity.

Question 9: A multiple linear regression model with $p = 3$ predictors is fit using $n = 100$ observations. Observation i has standardized residual $r_i = 2.4$ and leverage $h_{ii} = 0.25$.

Use

$$D_i = \frac{r_i^2}{p+1}\left(\frac{h_{ii}}{1-h_{ii}}\right).$$

$\frac{2.4^2}{4}\left(\frac{.25}{.75}\right)$

(Circle one answer) Which conclusion is most appropriate?

(A) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(B) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(C) $D_i = 0.48$. The observation is not influential because its leverage is less than 0.50.

(D) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(E) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(F) $D_i = 0.2$. The observation is not influential because its leverage is less than 0.50.

(G) Cook's distance cannot be calculated without knowing the observation's original residual.

Question 10:

(A) What is the formula for $\widehat{\beta}$?

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

(B) What is the conceptual difference between $\widehat{\beta}$ and β?

β is an unknown parameter of the population.

$\hat{\beta}$ is an estimated metric from the sample for our model

(C) Explain how least squares can be used to estimate regression coefficients. You do not need to derive anything, but you should be clear about how this is done conceptually.

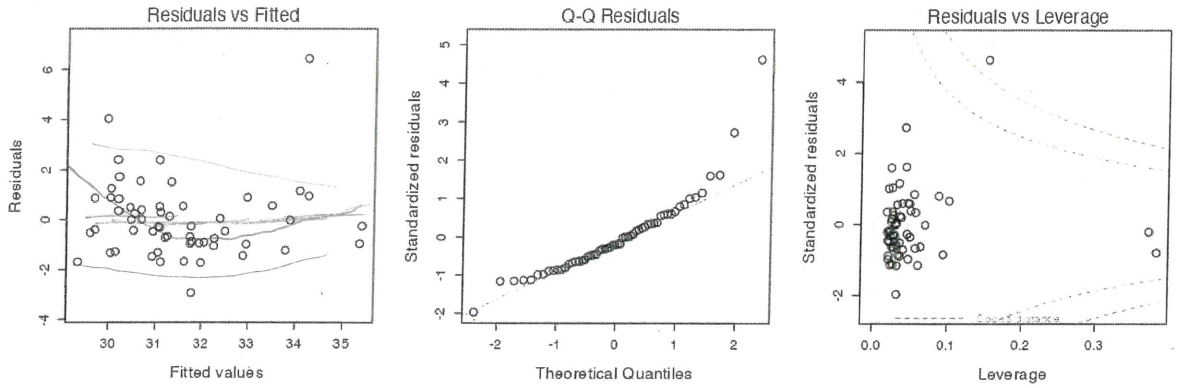
Least squares is used to find the possible coefficient that accounts for the largest possible portion of variation with the responses. Therefore leaving the smallest SSE.

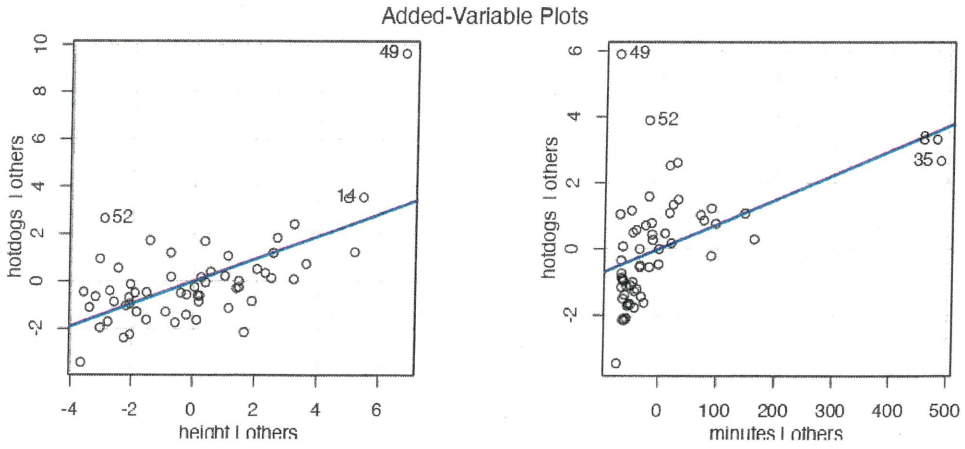
PART II

Some consider competitive eaters to be among the most inspiring individuals in modern times. We are interested in knowing what factors relate to a competitive eater's ability to consume. In a recent state fair, a contest was held where contestants competed to see who could eat the most hotdogs in 10 minutes. Data were collected on 58 amateur competitive eaters, most of whom were from the 4 largest counties in the state. For each eater, the total number of hotdogs consumed in a single 10 minute sitting was recorded. Each contestant's height in inches was also recorded, and each contestant was asked "How many minutes per week do you spend training for a contest?" The variables are `hotdogs`, `height`, and `minutes`. Model (1) is used for analysis. Use the plots and model fit output below to answer questions 11-13.

Model 1

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \texttt{minutes}_i + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{1}$$





Coefficient	Estimate	Std. Error	t value	$\Pr(>\|t\|)$
Intercept	-1.295392	5.467711	-0.237	0.813602
`height`	0.472892	0.080352	5.885	2.46e-07
`minutes`	0.007417	0.001871	3.963	0.000215

Other Summaries	Value
Breusch-Pagan p-value	0.071
Shapiro-Wilk p-value	2.43e-8
CV error	1.84623

Question 11: State the estimated value of β_1, and interpret its value in the context of the problem

$\hat{\beta}_1 = .472892$ We estimate that holding minutes constant there is an associated .473 of a hot dog increase for each additional inch in height of a contestant

Question 12: **(Circle all that apply)** Which of the following are true?

(A) The 95% confidence interval of β_2 does **not** contain 0

(B) The 95% confidence interval of β_2 does contain 0

(C) There is an 81.4% probability that $\beta_0 = 0$

(D) $\beta_1 = 0.4729$

Question 13: Given what you know, provide a brief assessment of how well the data satisfy each of the following multiple linear regression assumptions:

(A) The relationships between the explanatory and response variables are linear

Based on the Residuals vs Fitted values plot there appears to be some curvature. And looking at the minutes to response added variable plot the relationship doesn't seem very linear, so we have some concern about the linearity of the relationship between predictors & response

(B) There is independence between each observation

We don't know much about the contestants, but it doesn't seem extremely bad. given that hot dog eating seems like a more rural competition and most contestants are from the largest counties (which are also probably rural) we don't know if some contestants might be family or train together so we might have concerns on independence.

(C) The errors are normally distributed

The QQplot shows some departure from the line x=y And the shapiro-wilk test has an extremely small p-value, so we are somewhat concerned about the normality of the error distribution

(D) The errors have constant variance (homoscedasticity)

The breusch-pagan test has a p-value close to but not quite beyond the alpha level of .05, and the Fitted values vs resid plot doesn't show a strong funnel shape, so we are probably ok.

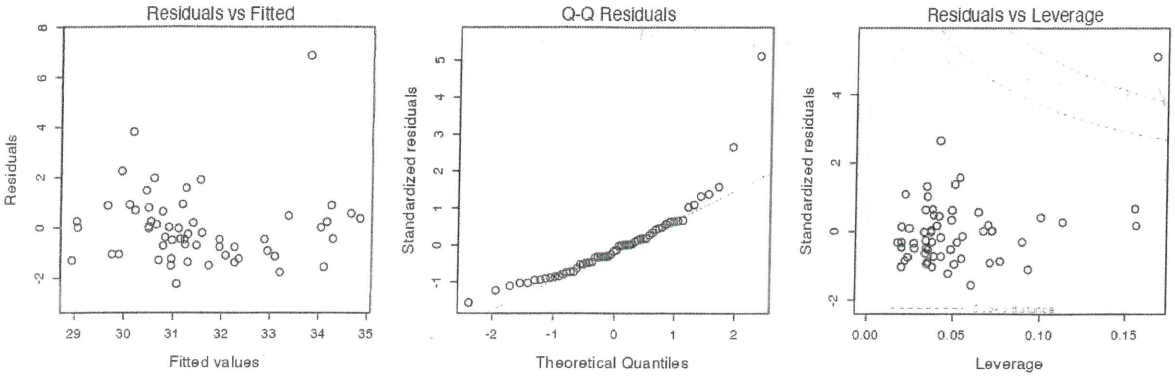
(E) There are no unduly influential points

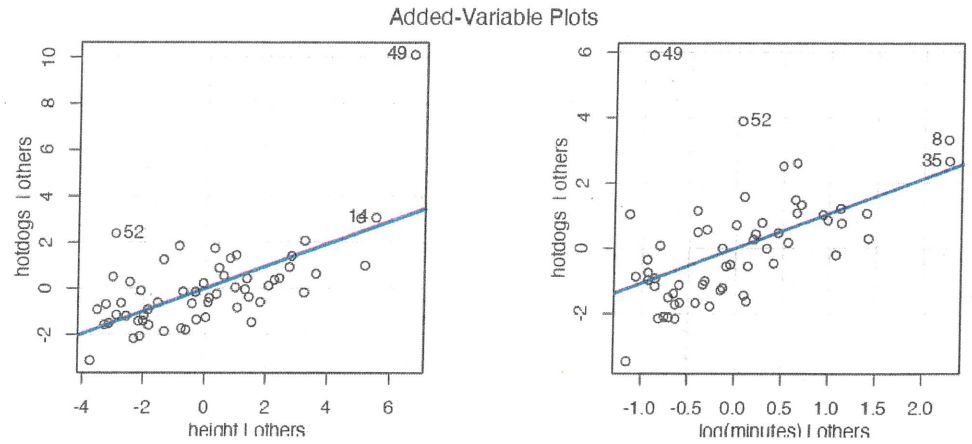
Based on the standardized resids vs leverage plot there are some points certainly with high leverage, beyond .3, but only 1 point that appears to have an extreme cook's distance & therefore unduly high influence.

Model 2

Model (2) below represents another possible multiple linear regression model for the same data. Here the variable `minutes` was log-transformed prior to fitting the model. Use model (2) and the output below to answer questions 14-16

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \log(\texttt{minutes}_i) + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{2}$$





Coefficient	Estimate	Std. Error	t value	$\Pr(>\|t\|)$
Intercept	-6.05589	5.43620	-1.114	0.27
`height`	0.48834	0.07821	6.244	6.48e-08
log(`minutes`)	1.06702	0.24079	4.431	4.51e-05

Other Summaries	Value
Breusch-Pagan p-value	0.028
Shapiro-Wilk p-value	3.43e-5
CV RMSE	1.66702

Question 14: A Breusch-Pagan test for model (1) results in a p-value of 0.071 while a Breusch-Pagan test for model (2) results in a p-value of 0.028. A researcher concludes that model (1) is clearly superior to model (2). How would you respond?

while it does have a larger p-value on that test the transformation clearly improved linearity between response & minutes.

There is more nuance in determining which model is better than just 1 test

model 2 also has a smaller CV-RMSE which is better.

Question 15: Explain what advantage(s) and disadvantage(s) model (2) provides over model (1)

smaller CV-RMSE, better linearity between predictors & response and larger though still significant shapiro-wilk p-value are advantages

disadvantages include the significant .071 breusch pagan test p-value

Question 16: State the estimated value of β_2 and provide an interpretation in context

The $\hat{\beta}_2 = 1.06702$

For every 1% increase in minutes trained per week there is an associated .01067 hot dog consumption increase.

Use the following for questions 17-19

Your best friend reveals that he will be competing in the hotdog eating contest at the next state fair. You know that he is 68 inches tall and that he trains 79 minutes per week. Note that log(79) = 4.369. Use this information and the fit for model (2) to answer the following questions.

Question 17: Use model (2) to calculate the predicted number of hotdogs your friend can consume in 10 minutes

$\widehat{hotdogs}_{friend} = \beta_0 + \beta_1(height) + \beta_2 log(minutes$

$hot\ dogs = -6.05589 + .48834(68) + 1.06702(4.369) = 31.81$ hot dogs

Question 18: To create a confidence or prediction interval for the expected number of hotdogs your friend can consume in 10 minutes, what is the correct number of degrees of freedom for the critical t^* used to calculate the margin of error?

(A) 55

(B) 57

(C) 56

(D) 48

$n-p-1 = 58-2-1 = 55$

Question 19: You're given that $s\sqrt{x^*(X^\top X)^{-1}x^*} = 0.204$, $s\sqrt{1 + x^*(X^\top X)^{-1}x^*} = 1.485$, and $t^*_{df,0.975} = 2.004$. Give a 95% *prediction* interval for the number of hotdogs your friend can consume in 10 minutes.

$95\%\ PI = estimate \pm t^*(s\sqrt{1+x^*(X^TX)^{-1}x^*})$

$31.81304 \pm 2.004(1.485)$

95% PI [28.8571, 34.78898]

Part III - Application

Questions 20-24 concern the following dataset:

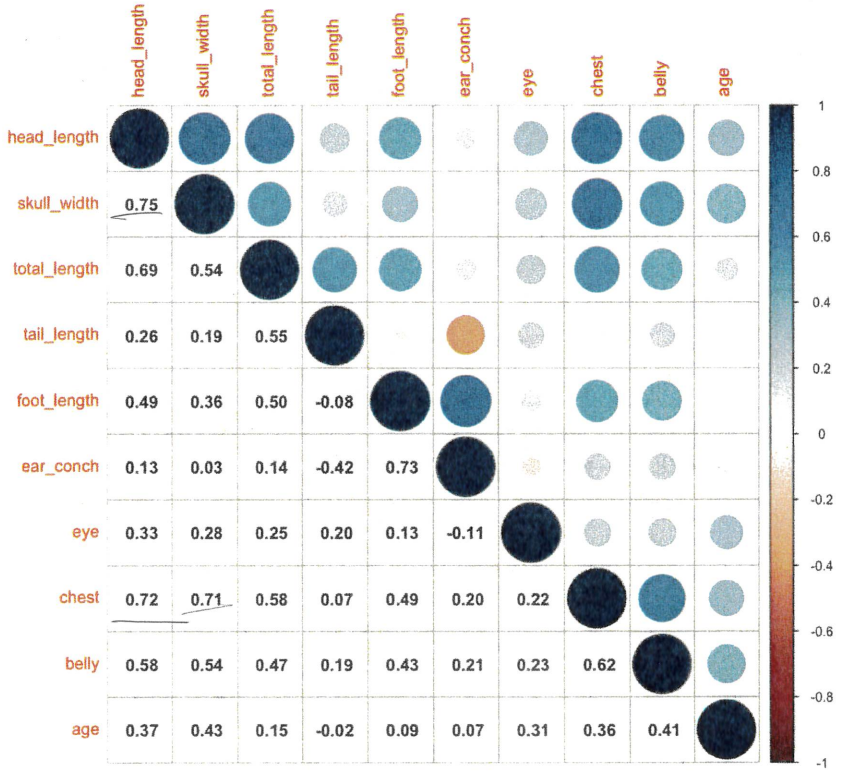
In 1995, Australian zoologists conducted a study of local possum populations. They trapped dozens of specimens, determined their ages, and took measurements of several physical attributes from each possum. The process of determining a wild animal's age can be challenging and imprecise, and usually requires a detailed analysis of the animal's teeth, skin, skeletal structure, or DNA. Among other questions, researchers were interested in whether a possum's age could be predicted using only easy to obtain physical measurements. For this analysis we will focus on the 59 male possums from the sample. The variables in the dataset are as follows (see below). A multiple linear regression model to predict age using all other variables was fit to these data, resulting in the following model summary and coefficient estimates (see below)

Variable	Description
head_length	in mm
skull_width	in mm
total_length	in cm
tail_length	in cm
foot_length	in mm
ear_conch	ear conch circumference (in mm)
eye	width of right eye (in mm)
chest	chest girth (in cm)
belly	belly girth (in cm)
age	in years

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -13.22377    7.34105  -1.801   0.0778 .
head_length   0.07077    0.10120   0.699   0.4877
skull_width   0.16559    0.10599   1.562   0.1247
total_length -0.04752    0.10167  -0.467   0.6423
tail_length  -0.05049    0.17030  -0.296   0.7681
foot_length  -0.21499    0.10733  -2.003   0.0507 .
ear_conch     0.15730    0.10231   1.538   0.1306
eye           0.43736    0.22549   1.940   0.0582 .
chest         0.01883    0.18154   0.104   0.9178
belly         0.20983    0.11580   1.812   0.0761 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.655 on 49 degrees of freedom
Multiple R-squared:  0.3538,    Adjusted R-squared:  0.2352
F-statistic: 2.982 on 9 and 49 DF,  p-value: 0.006552
```

A correlation plot (shown below) of these variables points towards some level of multicollinearity in the data. The maximum VIF among these variables is approximately 4 with an average VIF of approximately 3.



Question 20:

(a) Using the information from the summary table above, produce a 95% confidence interval for the coefficient on `skull_width`. The 0.975 critical value of a t-distribution with 49 degrees of freedom is 2.01. You may report your answer in the form $a \pm b * c$.

$.16559 \pm 2.01 \times 1.0599$

(b) A 95% confidence interval for $\beta_{\texttt{belly}}$, the coefficient on belly width, is (-0.023, 0.443). Interpret this interval within the context of the data.

We are 95% confident that the associated change of age in years is captured within the range of (-.023, .443) for every 1 cm increase in belly width

Question 21: Note from the model summary that while the overall F-test of model significance produces a p-value of 0.0065, none of the individual model coefficients are found to have a significant effect at the $\alpha = 0.05$ level. What do you think is driving this "contradiction"? (It's not actually a contradiction.)

It is probably multicollinearity, we have varios predictors that are highly correlated such as head length and skull width and head length and chest girth among others.

Question 22: Briefly describe how the "best subsets" algorithm using AIC as its criterion performs variable selection.

The best subsets algorithm using AIC is an exhaustive method that will test every possible model applying an AIC penalty to determine which models give the lowest RMSE.

Question 23: Best subsets using AIC as a criterion identified the following model with five variables:
`lm(age ~ skull_width + foot_length + ear_conch + eye + belly, data = possum)`
This model produces the following summary and coefficient estimates:

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -14.76661    4.97828  -2.966  0.00451 **
skull_width   0.20221    0.07990   2.531  0.01439 *
foot_length  -0.22936    0.09385  -2.444  0.01789 *
ear_conch     0.18239    0.08653   2.108  0.03979 *
eye           0.45001    0.21687   2.075  0.04285 *
belly         0.21405    0.10316   2.075  0.04287 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.609 on 53 degrees of freedom
Multiple R-squared:  0.3396,    Adjusted R-squared:  0.2773
F-statistic:  5.45 on 5 and 53 DF,  p-value: 0.0004056
```

Is this model guaranteed to have the highest adjusted-R^2 among all possible models.

(A) TRUE

(B) FALSE

Question 24: Suppose we want to compare the out-of-sample predictive performance of the full model and the model chosen by best subsets, and ultimately select the model with the smaller cross-validated RMSE.

Cross-validation using repeated random training-test splits can be structured around a for loop over some number of iterations (let's say 100), with some setup happening prior to the start of the loop, the bulk of the code within the loop, and some code after the loop has ended to process the results. To be clear, we **ARE NOT** using k-fold cross validation.

For each of the following steps in the cross-validation process, identify (by writing in the blank space) whether it occurs:

(A) prior to the for loop

(B) within the for loop

(C) after the for loop

(D) never in the CV process

C Average RMSE across all test-training splits and report

C Average the predictions and record their value

D Choose the model with the largest RMSE

C Choose the model with the smallest RMSE

B Compare the predictions from each model (that was fitted to training data) to the observed test data and compute RMSE

B Compute each model's AIC and BIC

D Fit each model using the testing data

B Fit each model using the training data

A Initialize vectors for each model that you will use to store their RMSEs

B Refit each model using the full data set and compare the coefficient estimates to the true values

A Split the data into test and training sets

B Use the models fitted to the training data to get predictions for the test data

END OF EXAM

STAT 330-001

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Asplund, Evan

765427658 scooby14 805-835-9313

Test: 641692 Form: A
Entry: 08:26pm 10/21/2026
Employee: Addison R.

Wadsworth, Spencer Gordon
Checked out:

SPECIAL INSTRUCTIONS

TC Calc
Notes: 3x5
Dictionary
Take a deep breath. Be careful to read instructions thoroughly. Answer questions completely.

Please keep this sheet **VISIBLE** on your desk at all times. When you are finished, *staple this sheet* to the front of your test booklet and return all materials to the Grading Table. You may use the space below for notes or scratch paper.

PLEASE TURN OFF YOUR CELL PHONE

NS

40142034

071

STAT 330 - Exam 1

Section 001 (Wadsworth), Section 002 (White)
October 16th - 21st, 2026

Instructions

- You may not refer to your notes, other people, or other resources. You can, however, use one 3 inch by 5 inch note card that you create (writing on both sides is allowed).
- Do not discuss any aspect of this exam with anyone except a Stat 330 TA or instructor until scores are posted.
- You are allowed to use a calculator for this exam.
- In Parts II and III, **make sure to clearly show your work for any calculations that you make and to give your answers in context of the data sets.** Place your final answer inside the box, if one is provided.
- There are 24 questions. Make sure you answer all of them.
- There is extra scratch paper at the end of the exam.

PART I

Question 1: (Circle one answer) Partial regression (or added-variable) plots illustrate the effect of adding an additional variable to the model after accounting for the variables that are already included in the model.

(A) TRUE

(B) FALSE

Question 2: (TRUE/FALSE) If the relationship between x and y is nonlinear, a simple linear regression model may still be feasible after transforming the data.

- TRUE
- FALSE

Question 3: (Circle one answer) When adding an additional variable to a linear regression model, the model's R^2 ...

(A) always increases or remains unchanged.

(B) always decreases or remains unchanged.

(C) may increase or decrease depending of the additional variable.

Question 4: (Circle all that apply) Which of the following statements describe possible effects of multicollinearity?

LINE

(A) It can cause the linearity assumption to be violated.

(B) It can cause estimated coefficients to have signs opposite those of their true population coefficients.

(C) It can result in larger standard errors, and thus greater uncertainty in estimates.

(D) It can cause the coefficient of determination (R^2) to become negative.

Question 5: (Circle all that apply) Which of the following are true statements about multiple linear regression (MLR)?

(A) BIC will usually favor a model with more predictors than the AIC criterion when performing model selection.

(B) Cook's distance can be used to identify influential observations.

(C) Transformations of the x's or y may help alleviate problems with linearity, equal variance, or normality.

(D) The least-squares estimate of β can be obtained by fitting a simple linear regression for each x_k only.

(E) The assumptions on the error variables (i.e., the ϵ_i) in a multiple linear regression model are exactly the same as they are for simple linear regression.

Question 6: (Circle one answer) What is the primary benefit of using cross-validation (CV)?

(A) CV reduces variance in the sampling distribution of regression coefficients by splitting the training and test data.

(B) CV gives a more realistic representation of model performance by predicting observations that were not used to fit the model.

(C) CV ensures model assumptions are met which eliminates a time-consuming step of the modeling process.

(D) CV is the most efficient way to calculate the shrunken coefficient estimates in LASSO and ridge regression models.

Question 7: (Circle all that apply) In datasets with high multicollinearity and/or redundant covariates, what advantages do LASSO and/or ridge regression have in comparison to the traditional ordinary least squares (OLS) approach to model fitting?

(A) LASSO can set coefficients to be exactly equal to zero, and thus serves as an automated approach to variable selection

(B) Their coefficient estimates are less biased than those of OLS

(C) They produce more accurate confidence intervals and hypothesis tests

(D) Ridge regression often handles multicollinearity well

(E) They can handle datasets where the number of predictors exceeds the number of observations

(F) They can take advantage of the bias-variance tradeoff to produce more stable and accurate predictions

Question 8: Why are LASSO and ridge regression sometimes referred to as "shrinkage" methods? Specifically, what is being shrunk and how is this accomplished?

They transform the data

λ is being enlarged in order to shrink outliers

$4/100 = 0.04$

Question 9: A multiple linear regression model with $p = 3$ predictors is fit using $n = 100$ observations. Observation i has standardized residual $r_i = 2.4$ and leverage $h_{ii} = 0.25$.

Use

$$D_i = \frac{r_i^2}{p+1}\left(\frac{h_{ii}}{1-h_{ii}}\right).$$

$= \left(\frac{2.4^2}{3+1}\right)\left(\frac{0.25}{1-0.25}\right) = (1.44)\left(\frac{1}{3}\right) = 0.48$

(Circle one answer) Which conclusion is most appropriate?

(A) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(B) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(C) $D_i = 0.48$. The observation is not influential because its leverage is less than 0.50.

(D) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(E) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(F) $D_i = 0.2$. The observation is not influential because its leverage is less than 0.50.

(G) Cook's distance cannot be calculated without knowing the observation's original residual.

Question 10:

(A) What is the formula for $\widehat{\beta}$?

$\hat{\beta} = (X^TX)^{-1}(X^TY)$

(B) What is the conceptual difference between $\widehat{\beta}$ and β?

β is unknown

$\hat{\beta}$ is our best estimate

(C) Explain how least squares can be used to estimate regression coefficients. You do not need to derive anything, but you should be clear about how this is done conceptually.

$e = Y - X\beta$

$SSE = (Y-X\beta)^T(Y-X\beta)$

$\hat{\beta} = (X^TX)^{-1}(X^TY)$

$\hat{Y} = X\hat{\beta} = HY$

$H = X(X^TX)^{-1}X^T$

$s^2 = SSE/(n-p-1)$

Least squares uses $(X^TX)^{-1}(X^TY)$ to estimate regression coefficients.

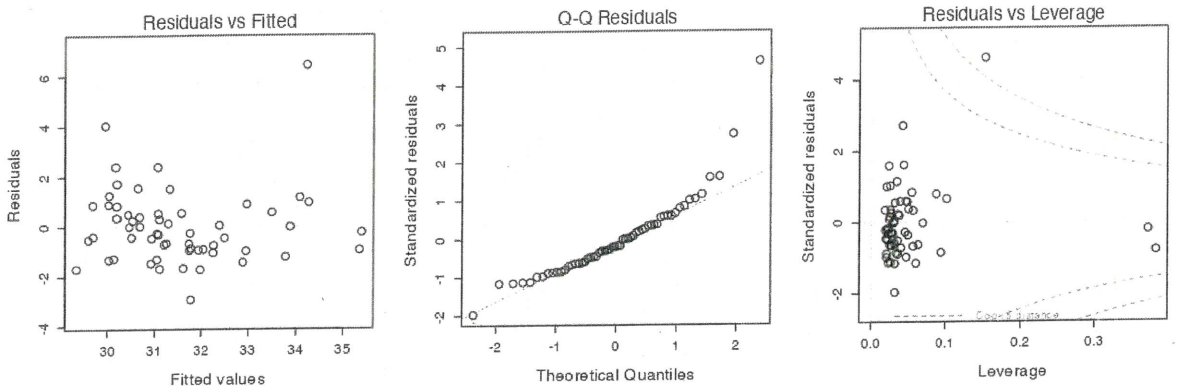
PART II

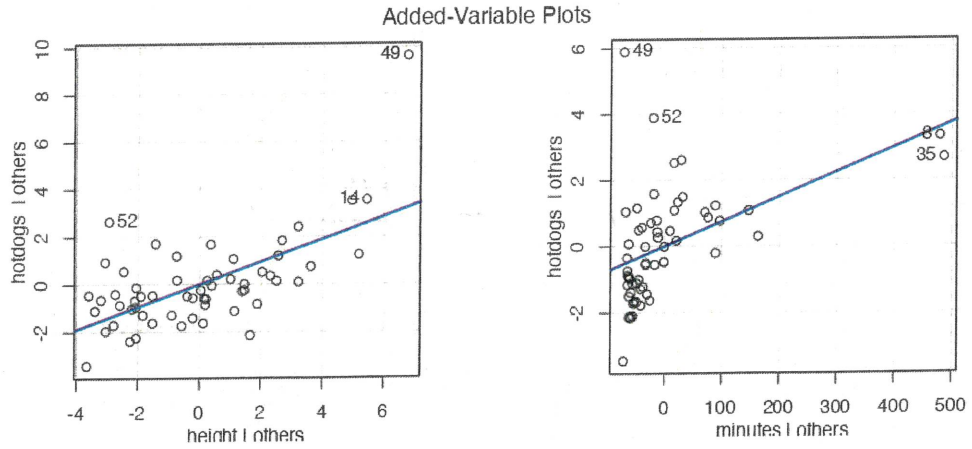
Some consider competitive eaters to be among the most inspiring individuals in modern times. We are interested in knowing what factors relate to a competitive eater's ability to consume. In a recent state fair, a contest was held where contestants competed to see who could eat the most hotdogs in 10 minutes. Data were collected on 58 amateur competitive eaters, most of whom were from the 4 largest counties in the state. For each eater, the total number of hotdogs consumed in a single 10 minute sitting was recorded. Each contestant's height in inches was also recorded, and each contestant was asked "How many minutes per week do you spend training for a contest?" The variables are `hotdogs`, `height`, and `minutes`. Model (1) is used for analysis. Use the plots and model fit output below to answer questions 11-13.

$n = 58$

Model 1

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \texttt{minutes}_i + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{1}$$





Coefficient	Estimate	Std. Error	t value	$\Pr(>\|t\|)$
Intercept	-1.295392	5.467711	-0.237	0.813602
`height`	0.472892	0.080352	5.885	2.46e-07
`minutes`	0.007417	0.001871	3.963	0.000215

Other Summaries	Value
Breusch-Pagan p-value	0.071
Shapiro-Wilk p-value	2.43e-8
CV error	1.84623

Question 11: State the estimated value of β_1, and interpret its value in the context of the problem

0.472892

Holding minutes constant, a 1-unit increase is associated with an estimated 0.472892 increase in average hot dogs.

Question 12: **(Circle all that apply)** Which of the following are true?

$\hat{\beta} \pm t^* SE$

$0.007417 \pm 3.963(0.001871)$

$\vee \quad \pm 0.006909603$

$(0.014327, 0.000507)$

(A) The 95% confidence interval of β_2 does **not** contain 0

(B) The 95% confidence interval of β_2 does contain 0

(C) There is an 81.4% probability that $\beta_0 = 0$

(D) $\beta_1 = 0.4729$

Question 13: Given what you know, provide a brief assessment of how well the data satisfy each of the following multiple linear regression assumptions:

(A) The relationships between the explanatory and response variables are linear

The residuals vs. fitted plot shows linearity is met

(B) There is independence between each observation

The methods for data collection show independence is met

(C) The errors are normally distributed

The Q-Q plot shows the estimated quantities tightly hugging the line, showing normality is met. The Shapiro-Wilk test value also significantly shows this with a very small p-value.

(D) The errors have constant variance (homoscedasticity)

The p-value for the Breusch Pagan test is greater than $\alpha = 0.05$, suggesting that homoscedasticity is not met.

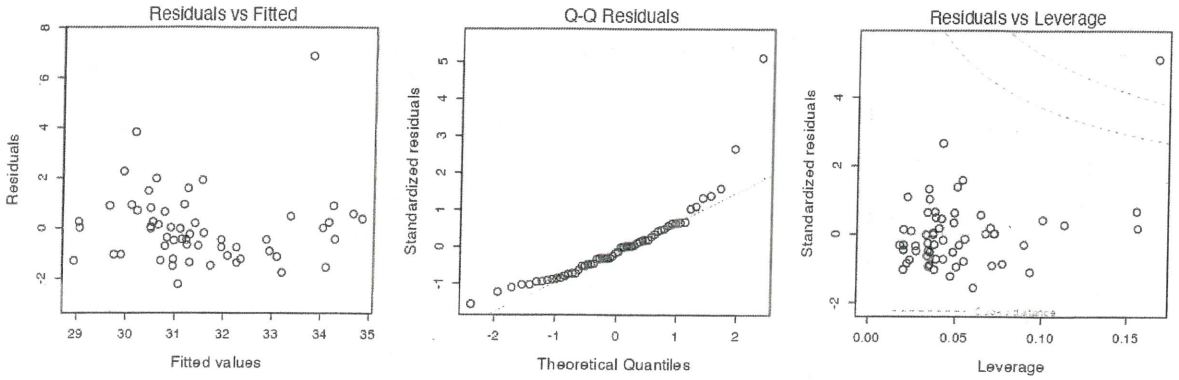
(E) There are no unduly influential points

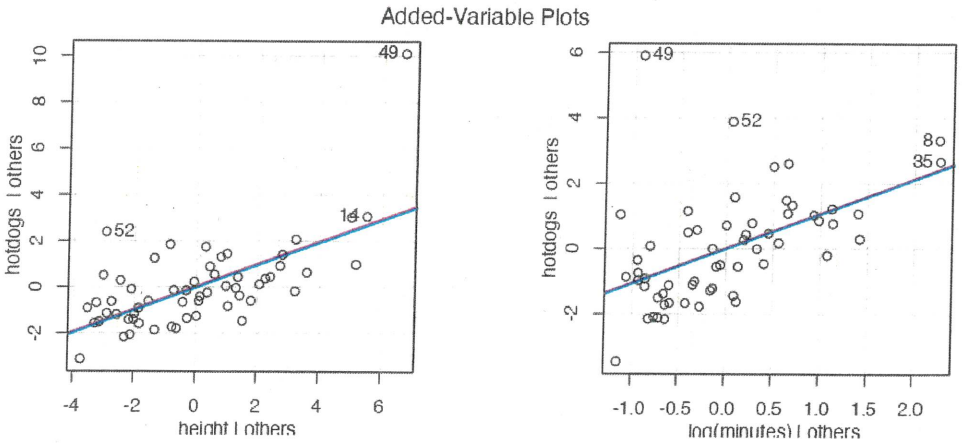
There is one unduly influential point found at 0 minutes with 52 hot dogs on the second AV plot that should be flagged.

Model 2

Model (2) below represents another possible multiple linear regression model for the same data. Here the variable `minutes` was log-transformed prior to fitting the model. Use model (2) and the output below to answer questions 14-16

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \log(\texttt{minutes}_i) + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{2}$$





Coefficient	Estimate	Std. Error	t value	$\Pr(>\|t\|)$
Intercept	-6.05589	5.43620	-1.114	0.27
`height`	0.48834	0.07821	6.244	6.48e-08
log(`minutes`)	1.06702	0.24079	4.431	4.51e-05

Other Summaries	Value
Breusch-Pagan p-value	0.028
Shapiro-Wilk p-value	3.43e-5
CV RMSE	1.66702

Question 14: A Breusch-Pagan test for model (1) results in a p-value of 0.071 while a Breusch-Pagan test for model (2) results in a p-value of 0.028. A researcher concludes that model (1) is clearly superior to model (2). How would you respond?

I would say that model (2) is actually the better model, where the p-value is less than $\alpha = 0.05$, meaning we reject H_0 & conclude that at least one predictor $\neq 0$.

Question 15: Explain what advantage(s) and disadvantage(s) model (2) provides over model (1)

Advantages include: outliers or unduly influential points are shrunk to a more normal level comparable to the rest of the data; a more reliable confidence interval.
Disadvantages include: log(minutes) can overinflate its influence compared to height.

Question 16: State the estimated value of β_2 and provide an interpretation in context

$\beta_2 = 1.06702$

Holding height constant, a 1-unit increase is associated with an estimated 1.06702 increase in average hot dogs.

Use the following for questions 17-19

Your best friend reveals that he will be competing in the hotdog eating contest at the next state fair. You know that he is 68 inches tall and that he trains 79 minutes per week. Note that log(79) = 4.369. Use this information and the fit for model (2) to answer the following questions.

Question 17: Use model (2) to calculate the predicted number of hotdogs your friend can consume in 10 minutes

mins height

$\log(10) \sim x$: $100(e^{\beta \cdot 1}) = 100(e^{0.48834 \cdot 1})$ 60 hot dogs

$= 59.9499 \approx$

Question 18: To create a confidence or prediction interval for the expected number of hotdogs your friend can consume in 10 minutes, what is the correct number of degrees of freedom for the critical t^* used to calculate the margin of error?

$df = n - p - 1$ β_1

$58 - p - 1$

(A) 55

(B) 57

(C) 56

(D) 48

Question 19: You're given that $s\sqrt{x^*(X^\top X)^{-1}x^*} = 0.204$, $s\sqrt{1 + x^*(X^\top X)^{-1}x^*} = 1.485$, and $t^*_{df,0.975} = 2.004$. Give a 95% *prediction* interval for the number of hotdogs your friend can consume in 10 minutes.

$PI = \hat{Y}^* \pm t^* SE$

$= 60. \pm (2.004)(1.485)$

$= (178.5564, -178.5564)$

Part III - Application

Questions 20-24 concern the following dataset:

In 1995, Australian zoologists conducted a study of local possum populations. They trapped dozens of specimens, determined their ages, and took measurements of several physical attributes from each possum. The process of determining a wild animal's age can be challenging and imprecise, and usually requires a detailed analysis of the animal's teeth, skin, skeletal structure, or DNA. Among other questions, researchers were interested in whether a possum's age could be predicted using only easy to obtain physical measurements. For this analysis we will focus on the 59 male possums from the sample. The variables in the dataset are as follows (see below). A multiple linear regression model to predict age using all other variables was fit to these data, resulting in the following model summary and coefficient estimates (see below)

Variable	Description
head_length	in mm
skull_width	in mm
total_length	in cm
tail_length	in cm
foot_length	in mm
ear_conch	ear conch circumference (in mm)
eye	width of right eye (in mm)
chest	chest girth (in cm)
belly	belly girth (in cm)
age	in years

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -13.22377    7.34105  -1.801   0.0778 .
head_length   0.07077    0.10120   0.699   0.4877
skull_width   0.16559    0.10599   1.562   0.1247
total_length -0.04752    0.10167  -0.467   0.6423
tail_length  -0.05049    0.17030  -0.296   0.7681
foot_length  -0.21499    0.10733  -2.003   0.0507 .
ear_conch     0.15730    0.10231   1.538   0.1306
eye           0.43736    0.22549   1.940   0.0582 .
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.655 on 49 degrees of freedom
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F-statistic: 2.982 on 9 and 49 DF,  p-value: 0.006552
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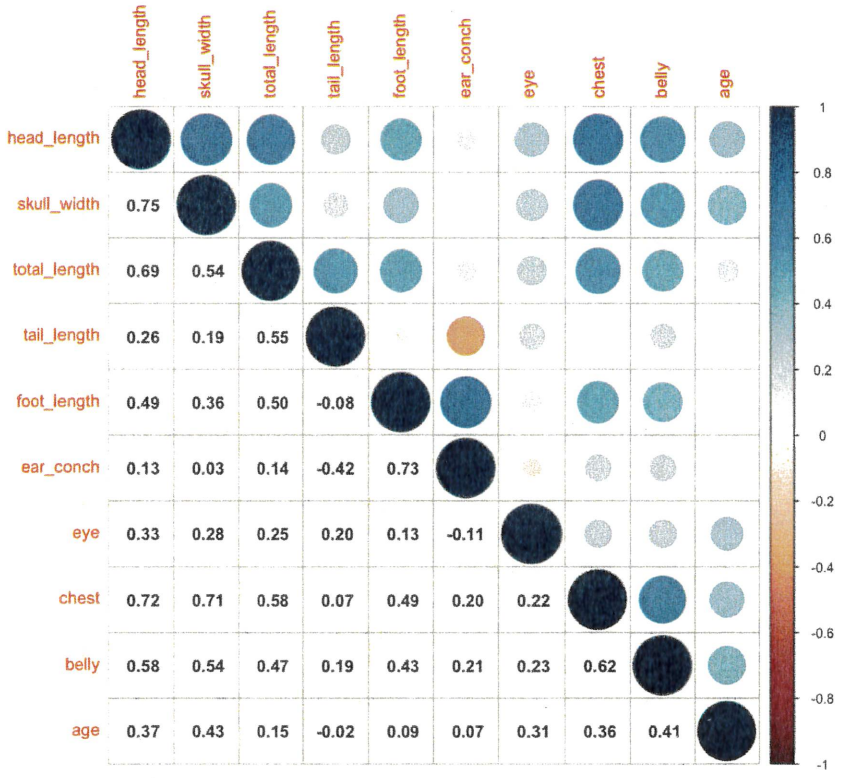
A correlation plot (shown below) of these variables points towards some level of multicollinearity in the data. The maximum VIF among these variables is approximately 4 with an average VIF of approximately 3.



Question 20:

(a) Using the information from the summary table above, produce a 95% confidence interval for the coefficient on `skull_width`. The 0.975 critical value of a t-distribution with 49 degrees of freedom is 2.01. You may report your answer in the form $a \pm b * c$.

$0.16559 \pm 2.01(0.10599)$

0.16559 ± 0.2130399

$(0.37863, -0.04745)$

(b) A 95% confidence interval for $\beta_{\texttt{belly}}$, the coefficient on belly width, is (-0.023, 0.443). Interpret this interval within the context of the data.

Seeing that the confidence interval contains 0, we conclude p-value > 0.05, meaning we fail to reject H_0. This is not proof that $\beta_{belly} = 0$.

Question 21: Note from the model summary that while the overall F-test of model significance produces a p-value of 0.0065, none of the individual model coefficients are found to have a significant effect at the $\alpha = 0.05$ level. What do you think is driving this "contradiction"? (It's not actually a contradiction.)

With a p-value of 0.0065, we reject the null hypothesis of the F-test & conclude that at least one β (predictor) slope $\neq 0$.

This "contradiction" is rooted in the fact that our F-test does not indicate which predictor has a non-zero slope.

Question 22: Briefly describe how the "best subsets" algorithm using AIC as its criterion performs variable selection.

The "best subsets" algorithm using AIC chooses the X_i that is the best fit by calculating each variable's influence & choosing the one that impacts the dataset the most.

Question 23: Best subsets using AIC as a criterion identified the following model with five variables:
`lm(age ~ skull_width + foot_length + ear_conch + eye + belly, data = possum)`
This model produces the following summary and coefficient estimates:

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -14.76661    4.97828  -2.966  0.00451 **
skull_width   0.20221    0.07990   2.531  0.01439 *
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Is this model guaranteed to have the highest adjusted-R^2 among all possible models.

(A) TRUE

(B) FALSE

Question 24: Suppose we want to compare the out-of-sample predictive performance of the full model and the model chosen by best subsets, and ultimately select the model with the smaller cross-validated RMSE.

Cross-validation using repeated random training-test splits can be structured around a for loop over some number of iterations (let's say 100), with some setup happening prior to the start of the loop, the bulk of the code within the loop, and some code after the loop has ended to process the results. To be clear, we **ARE NOT** using k-fold cross validation.

For each of the following steps in the cross-validation process, identify (by writing in the blank space) whether it occurs:

(A) prior to the for loop

(B) within the for loop

(C) after the for loop

(D) never in the CV process

C Average RMSE across all test-training splits and report

C Average the predictions and record their value

D Choose the model with the largest RMSE

C Choose the model with the smallest RMSE

A Compare the predictions from each model (that was fitted to training data) to the observed test data and compute RMSE

A Compute each model's AIC and BIC

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A Initialize vectors for each model that you will use to store their RMSEs

B Refit each model using the full data set and compare the coefficient estimates to the true values

A Split the data into test and training sets

C Use the models fitted to the training data to get predictions for the test data

END OF EXAM